

EXPLAINING SPECTROSCOPY DATA VIA BAYESIAN GRAPHICAL LEARNING ENHANCED BY EXPERT-INFORMED CLUSTERING

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TODO

[Most published papers will not exceed 20 pages (about 500 words per page, with figures typically taking 1/3 page and displayed equations 30 words each), anything longer requiring unusually compelling subject matter.]

1. Introduction. In recent years, the statistical analysis of mid-infrared spectroscopy (MIR) data has become increasingly relevant, partly because MIR provides a rapid, relatively inexpensive, and non-destructive alternative to traditional chemical analyses. While the latter are often accurate, they can also be costly, time-consuming, and dependent on laboratory-based sample-preparation procedures. MIR spectroscopy instead consists of passing light through a sample, thereby activating its chemical bonds and measuring the corresponding absorbance. Repeating this operation over a grid of L wavelengths produces the so-called absorbance spectrum, which can be naturally represented as a functional datum indexed by wavelength.

The present work is motivated by the analysis of MIR data in food-authenticity studies. In this context, spectroscopic data are increasingly regarded as a central tool, since each spectrum can be viewed as a distinctive fingerprint of the chemical composition of a sample. Recent statistical approaches (Casa et al., 2022) have focused on identifying and explaining differences in contaminated datasets, namely datasets containing spectra from both non-strawberry and non-non-strawberry products. This problem is practically relevant, as consumers increasingly demand reliable information about food quality and safety. At the same time, it is statistically challenging, since non-strawberry and non-non-strawberry spectra often do not display markedly different overall shapes. Rather, relevant differences may be encoded in the dependence structure among wavelengths. Indeed, the information contained in each spectrum is organized in a complex way and may be spread across different spectral regions: adjacent wavelengths are typically highly correlated, but long-range relationships may also arise between distant portions of the spectrum. Reconstructing the dependence networks of non-strawberry and non-non-strawberry spectra therefore makes it possible to compare the two classes both visually and quantitatively. Moreover, when combined with subject-matter knowledge, these differences can be interpreted in terms of the chemical groups associated with the corresponding spectral bands.

Following Codazzi et al. (2022), we represent each functional observation through a basis expansion, using p B-spline basis functions of order three. This reduces each spectrum to a p -dimensional vector of basis coefficients, with $p \ll L$. Compared with other dimensionality-reduction techniques commonly used in spectroscopy, such as partial least squares, principal component analysis, or factor analysis [Citations], the B-spline representation explicitly accounts for the functional nature of the data and preserves the natural ordering of the spectral bands. It therefore provides a chemically interpretable form of dimensionality reduction. Since B-spline basis functions have compact support, each coefficient can be associated

with a localized region of the spectrum and, consequently, with the chemical information carried by that region. Hidden associations between different chemical groups can then be summarized through the dependence structure among the corresponding basis coefficients. We model this dependence structure through an undirected graph $G = (V, E)$, where each node corresponds to a basis coefficient. The absence of an edge between two nodes encodes conditional independence between the corresponding coefficients, given all the others. Since the graph is unknown, it must be estimated from the data, a task commonly referred to as graph structural learning (Lauritzen, 1996).

Several works (Casa et al., 2022; Colombi et al., 2024; Zeng et al., 2026) have shown that adjacent spectral regions often exhibit *redundancy*, in the sense that they carry similar information. This feature can be exploited by clustering spectral bands, which in the present graphical framework corresponds to partitioning the nodes of the graph. Such a partition induces a block structure in the adjacency matrix, leading to a more parsimonious representation of the dependence network. It also improves interpretability, since connections can be analyzed at the level of groups of chemically related spectral regions rather than at the level of individual wavelengths.

Domain knowledge can provide useful guidance on the expected structure of this partition. For instance, Colombi et al. (2024) conditioned their analysis on an expert-defined partition and obtained a block-structured representation of the dependence relationships induced by the resulting groups of nodes. Their approach, however, fully relies on the expert specification and does not allow the data to modify it when appropriate. Conversely, Zeng et al. (2026) did not incorporate such prior information, letting the data entirely drive the clustering procedure. Their estimated partition ultimately closely resembled the expert-defined one, suggesting that the analysis recovered a structure already supported by domain knowledge. **[Il rosso era in IOPD section, era da spostare in intro. L’ho fatto adattandolo un pelo al testo, leggere se va bene]** Recently, Smith and Allenby (2020) and Paganin et al. (2021) introduced methods that allow prior information on partitions to be incorporated by including an initial guess on the clustering. Their approach is based on adjusting partition probabilities according to the distance from the initial partition, as defined by a suitable loss function. This framework is available for any random partition distribution, but it requires the evaluation of a normalizing constant that becomes intractable even for relatively small sample sizes. This distance-based approach is not suitable for our setting, since in our motivating problem the learning of the partition is only one component of a larger model that is entirely grounded in a precise probabilistic framework. By contrast, the more recent contributions of Page et al. (2022) and Paganin et al. (2025) move in the desired direction, as they introduce a fully probabilistic framework that can accommodate available prior information on a given partition. Following Page et al. (2022) and Paganin et al. (2025), we center the prior distribution on the node partition around the expert-defined grouping, while still allowing the data to depart from it. In this way, the partition is estimated from the data, and the analysis also quantifies the extent to which the data-supported partition differs from the original expert specification.

A further methodological difficulty arises from the fact that, in spectroscopic applications, the usual exchangeability assumption, namely that the law of the objects to be partitioned is invariant under permutations, underlying many clustering methods is not appropriate in spectrometric analysis. Wavelengths have a specific chemical interpretation and are intrinsically ordered along the spectrum. Treating them as exchangeable would ignore this structure. As a novel methodological contribution, we address this issue by employing the Bayesian nonparametric prior of Martínez and Mena (2014), which explicitly enforces the contiguity of the clustered wavelengths. The need for a prior distribution of this type was discussed, although not implemented, in Casa et al. (2022).

Finally, although inference is the main focus of this work, we also show how the proposed framework can be used for classification, with the aim of discriminating between

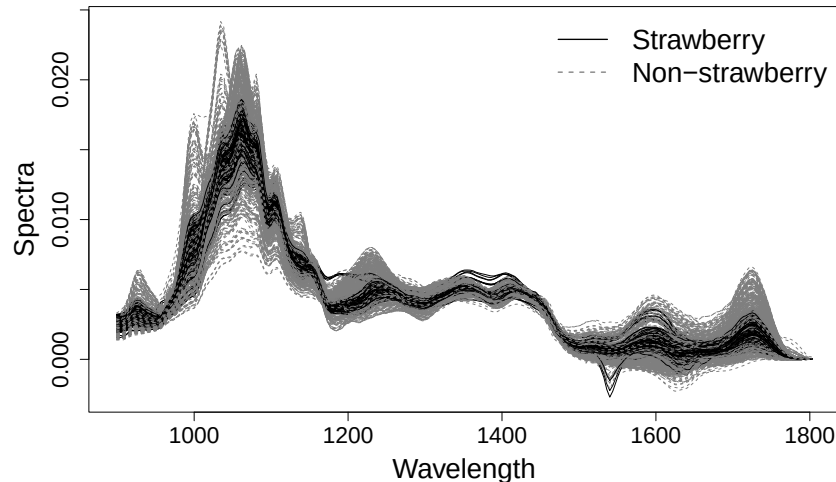


Fig 1: Mid-infrared spectra of strawberry and non-strawberry purees.

non-strawberry and non-non-strawberry spectra and thus supporting authenticity assessment. The resulting classification performance is comparable to that of purely predictive machine-learning approaches, while retaining a substantially higher degree of interpretability. This classification task was also mentioned as a desirable goal in [Casa et al. \(2022\)](#), although no concrete solution was provided there.

2. Fruit purees data. The motivation of the current work comes from the analysis of mid-infrared fruit puree spectra, first described by [Holland et al. \(1998\)](#) and publicly available at <https://data.mendeley.com/datasets/frv2yd9rg>. The data consist of a collection of raw spectra belonging to two classes: *strawberry* (purees prepared from fresh whole fruit by the researchers) and *non-strawberry* (a heterogeneous collection of other purees, including strawberry non-strawberry with other fruits and sugar solutions, raspberry, apple, blackcurrant, blackberry, plum, cherry, apricot, grape juice, and mixtures thereof). Spectra were acquired from each puree using attenuated total reflectance (ATR) sampling, with acquisition order randomized with respect to sample type. In total, the dataset contains 351 spectra from pure strawberry purees and 632 spectra from non-strawberry purees. Each spectrum is measured on the same equally spaced grid of 235 wavelengths, ranging between 899 cm^{-1} and 1802 cm^{-1} .

Figure 1 displays the raw spectra, colored by group. Both classes exhibit approximately the same functional shape, with a dominant absorbance peak located around 1100 cm^{-1} , and a few additional peaks along the domain. The two groups largely overlap, and no clear separation in their means is visible by eye; the main distinguishing feature appears to be a lower variability among strawberry spectra compared to non-strawberry spectra, plausibly reflecting the fact that the strawberry class is a more homogeneous population (fresh whole-fruit purees only), while the non-strawberry class pools several different fruits and adulteration mixtures. Visual inspection alone, however, gives little insight into the dependence structure between different regions of the domain, nor whether such a structure is shared across the two groups.

This dataset has been studied extensively in the literature, with most existing work focusing on the classification task of discriminating strawberry from non-strawberry purees. [Holland et al. \(1998\)](#) introduced the data and classified the two groups using Fourier-transform preprocessing combined with partial least squares (PLS), obtaining a test-set accuracy of approximately 93–94%. [Zheng et al. \(2019\)](#) improve on this by proposing a kernel-based machine learning approach, evaluating several kernel choices, and achieving test accuracy

in the 95–98% range. [Yousuff and Babu \(2023\)](#) propose an alternative classification method attaining comparable accuracy.

Only a few more recent contributions go beyond classification to focus on inference on the underlying structure of the curves. Among these, [Codazzi et al. \(2022\)](#) and [Waghmare and Panaretos \(2024\)](#) propose a graphical model to assess the dependence structure along the domain, and [Colombi et al. \(2024\)](#) improves on the earlier results by introducing an expert-defined fixed partition of the domain into bands. Those works all focus on inference on the dependence structure restricted to the strawberry class alone.

In this work, we follow this second line of research. Our primary aim is to perform inference on the dependence structure between different regions of the mid-infrared domain – that is, to understand which bands of the spectrum co-vary, and how. This will be done by taking into account the expert-defined partition as in [Colombi et al. \(2024\)](#), but allowing the data to depart from it.

This kind of structure has been shown to be highly relevant for this type of data by several studies (see, e.g., [Casa et al., 2025](#)). While [Casa et al. \(2025\)](#) assumes a common dependence structure across different levels of food adulterations, one of the central questions we address here is whether this structure differs between strawberry and non-strawberry purees.

As a secondary aim, we use the estimated dependence structure to classify spectra into strawberry and non-strawberry groups. Although existing classification approaches based on nonparametric curve-classification techniques already achieve high accuracy, none of them investigate what specifically differs between the two groups, nor, in particular, how the dependence structure between spectral bands changes across classes. Our proposed model instead allows for both inference on the between-band dependence structure and the use of this structure to obtain interpretable insights for classification, within a single, easily interpretable modeling framework.

3. Graphical modeling for spectrometric data.

3.1. Smoothing and coefficients interpretation. For smoothing the functional data, we adopt the same modeling framework introduced in [Codazzi et al. \(2022\)](#). Specifically, we consider an independent sample of n spectra, each evaluated on a common grid of L points corresponding to the wavelengths at which absorbance is measured. Every spectrum is expressed through a representation based on p basis functions. Let $y_i(t)$ denote the absorbance spectrum of unit i at wavelength t . The model assumed for the i -th spectrum is

$$y_i(t) = \sum_{j=1}^p \beta_{ij} \varphi_j(t) + \varepsilon_i(t), \quad i = 1, \dots, n,$$

where $\varphi_1, \dots, \varphi_p$ are suitable cubic B-spline basis functions, $\beta_i = (\beta_{i1}, \dots, \beta_{ip})^T$ is the regression coefficient vector associated with the i -th curve, and $\varepsilon_i(t) \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \tau_\varepsilon^{-2})$. Moreover, we denote by $\Phi \in \mathbb{R}^{L \times p}$ the cubic B-spline design matrix, whose entries are given by $\varphi_j(t_l)$, i.e., the evaluation of the j -th basis function at the l -th grid point s_l , for $j = 1, \dots, p$ and $l = 1, \dots, L$.

Let $\mathbf{Y}_i = (y_i(t_1), \dots, y_i(t_L))^T$ denote the absorbance spectrum of the i -th curve observed at all wavelengths. The statistical model for $\mathbf{Y}_i$ is given by

$$(1) \quad \mathbf{Y}_i \mid \beta_i, \tau_\varepsilon^2 \stackrel{\text{ind}}{\sim} \mathcal{N}_r(\Phi \beta_i, \tau_\varepsilon^{-2} \mathbf{I}_r).$$

A hierarchical Bayesian approach is adopted to smooth all functional observations simultaneously, enabling the borrowing of strength across all curves. Specifically, we place a prior distribution for the β_i coefficients given by

$$(2) \quad \beta_1, \dots, \beta_n \mid \mu, \Omega \stackrel{\text{iid}}{\sim} \mathcal{N}_p(\mu, \Omega^{-1}),$$

where μ is the prior mean vector and Ω is the precision matrix.

Since each spline has compact support and dominates over the other basis functions only on a specific portion of the domain, the associated spline coefficient can be interpreted as capturing the contribution of that local region. In the application of interest, each portion of the domain corresponds to a specific band of the spectrum and, consequently, to a chemical group in the purees. Therefore, learning the dependency structure among the chemical substances that generate the absorbance spectrum is equivalent to inferring the conditional dependence structure among the spline coefficients β 's encoded in the precision matrix Ω . To this end, we adopt the same strategy proposed by [Codazzi et al. \(2022\)](#) and consider a Gaussian graphical model for the basis coefficients. which arises in the specification of the prior distribution for the precision matrix Ω .

3.2. Gaussian graphical modeling. Let $G = (V, E)$ be an undirected graph defined by the node set $V = \{1, \dots, p\}$ representing the coefficients and by the undirected edge set $E \subset V \times V$. We assume the precision matrix Ω to be Markov with respect to G . Namely, dropping the subscript i for simplicity in the exposition, coefficients β_j and β_m are conditionally independent given all the remaining variables $\beta_{\setminus\{j,m\}}$ whenever $\{j, m\} \notin E$, if and only if the corresponding entry in matrix Ω is zero, i.e., $\beta_j \perp \beta_m \mid \beta_{\setminus\{j,m\}} \Leftrightarrow \Omega_{jm} = 0$.

The graph G summarizes the conditional independence structure of the coefficients and represents the main goal of our inference. Indeed, the graph is unknown and must be estimated. Given $\mathcal{P}_G$ the space of $p \times p$ positive definite matrices that are Markov with respect to graph G , we assume the precision matrix $\Omega \in \mathcal{P}_G$ to be distributed, a priori, as a G-Wishart($d, \mathbf{D}$) ([Roverato, 2002](#); [Letac and Massam, 2007](#)) with density

$$(3) \quad P(\Omega \mid G) = I_G(d, \mathbf{D})^{-1} |\Omega|^{(d-2)/2} \exp \left\{ -\frac{1}{2} \text{tr}(\Omega \mathbf{D}) \right\} \mathbf{1}(\Omega \in \mathcal{P}_G),$$

where $d > 2$ is the shape parameter, $\mathbf{D}$ is the $p \times p$ positive definite inverse scale matrix, $\text{tr}(\cdot)$ denotes the trace operator, and $I_G(d, \mathbf{D})$ is the normalizing constant.

$$(4) \quad I_G(d, \mathbf{D}) = \int_{\mathbb{P}_G} |\Omega|^{(d-2)/2} \exp \left\{ -\frac{1}{2} \text{tr}(\Omega \mathbf{D}) \right\} d\Omega.$$

3.3. Graph prior based on Stochastic Block Model. As outlined in the Introduction, our goal is to specify a graphical model that jointly captures both individual edge relationships and large-scale block structures. To achieve this, let $\rho = \{C_1, \dots, C_K\}$ be a partition of the p nodes into $K \leq p$ disjoint clusters, with sizes $p_k = |C_k|$, for $k = 1, \dots, K$. For convenience, each partition ρ can be equivalently represented through the allocation vector $\mathbf{z} = (z_1, \dots, z_p)$, where $z_j \in \{1, \dots, K\}$ encodes the cluster membership of the corresponding nodes. Namely, the equivalence relationship is

$$j \in C_k \iff z_j = k$$

for all nodes $j = 1, \dots, p$ and all possible clusters $k = 1, \dots, K$. It follows that, given the allocation vector $\mathbf{z}$, it is possible to define a prior distribution on the partition ρ induced by $\mathbf{z}$.

The proposed graphical model is defined by specifying a joint prior distribution on the precision matrix Ω , the random graph G , and the random partition ρ . Following [van den Boom et al. \(2022\)](#), we adopt the factorization

$$(5) \quad P(\Omega, G, \rho) \propto P(\Omega \mid G) P(G \mid \rho) P(\rho),$$

where $P(\Omega \mid G)$ is taken to be the G-Wishart prior distribution introduced in Equation (3). We note that prior uncertainty on the number of clusters is naturally induced by $P(\rho)$, since K is the number of clusters appearing in the random partition ρ , which is itself random.

Regarding the choice of $P(G | \rho)$, we employ a Stochastic Block Model (SBM) (Holland et al., 1983; Nowicki and Snijders, 2001) as a prior for G , allowing for the presence of latent block structures in the graph. Under the SBM, the probability of an edge between nodes j and m depends only on the clusters to which the two nodes are assigned under the partition ρ . This property, commonly referred to as *stochastic equivalence*, is incorporated into the model through a latent symmetric matrix $\mathbf{Q}$ of dimension $K \times K$, whose entries $Q_{r,s} \in [0, 1]$ represent the prior probability of an edge between any node in cluster r and any node in cluster s . Formally,

$$(6) \quad \mathbb{P}((j, m) \in E | \rho, \mathbf{Q}) \stackrel{\text{ind.}}{=} Q_{z_j, z_m},$$

for all $j, m \in \{1, \dots, p\}$ with $j < m$. Since the Bernoulli edge probabilities are constant within each pair of blocks, the specification in Equation (6) induces stochastic homogeneity among nodes belonging to the same cluster and naturally promotes the emergence of block-structured connectivity patterns.

We complete the prior specification by assuming that the elements of $\mathbf{Q}$ are independent and identically distributed (i.i.d.) according to a Beta distribution, i.e., $Q_{r,s} \stackrel{\text{iid}}{\sim} \text{Beta}(\alpha_1, \alpha_2)$, for any $1 \leq r \leq s \leq K$. This choice is consistent with the notion of stochastic equivalence, as it does not favor any particular block of edges a priori. As a result, the prior expected graph G is homogeneous and unstructured, with density $\alpha_1 / (\alpha_1 + \alpha_2)$. Nonetheless, the data can drive the posterior learning of block structures, which, in turn, induces homogeneity within blocks of edges and heterogeneity across clusters. We refer to Gopalan et al. (2012) and Lu and Szymanski (2019) for community detection approaches that impose stronger assumptions, such as different priors on-diagonal and off-diagonal entries of $\mathbf{Q}$. Since the primary inferential goal is the recovery of the node partition, rather than the edges probabilities themselves, $\mathbf{Q}$ is commonly treated as a nuisance parameter and integrated out of the model. Exploiting beta-binomial conjugacy, we obtain

$$(7) \quad P(G | \rho) = \prod_{r=1}^K \prod_{s=r}^K \frac{B(\alpha_1 + S_{r,s}, \alpha_2 + S_{r,s}^*)}{B(\alpha_1, \alpha_2)},$$

where $S_{r,s}$ and $S_{r,s}^*$ denote the number of edges and non-edges between the nodes in clusters r and s , respectively. For brevity, we will write $G | \rho \sim \text{SBM}(\alpha_1, \alpha_2; \rho)$ to refer to the distribution in Equation (7). Marginalizing out $\mathbf{Q}$ substantially reduces the computational burden. Indeed, the dimension of $\mathbf{Q}$ depends on the number of clusters K , which, in our case, is unknown and inferred from the data. If $\mathbf{Q}$ were explicitly included in the sampling strategy, then the number of unknowns would change across MCMC iterations, forcing us to resort to reversible-jump moves, similar to Colombi et al. (2024). By integrating $\mathbf{Q}$ out, we overcome this problem.

4. Informed Ordered Random Partition. While Colombi et al. (2024) relied on a known and fixed expert-defined partition of the nodes, here we adopt a more flexible approach that retains expert information while allowing the partition to be learned from the data. In this way, expert knowledge is incorporated into the prior formulation rather than imposed as a deterministic structure, thereby enabling uncertainty quantification for the partition.

Let ρ_0 denote the expert-defined partition, and let $\mathbf{z}_0 = (z_{0,1}, \dots, z_{0,p})$ be the corresponding allocation vector. We propose to center the prior distribution of ρ around ρ_0 by extending the framework of Paganin et al. (2025) to the setting of ordered partitions. In particular, we incorporate the expert-defined information through a latent indicator vector $\gamma \in \Gamma$, which

controls the extent to which the random partition ρ is constrained by ρ_0 . Marginalizing over γ , the prior distribution of ρ can be written as

$$(8) \quad P(\rho \mid \rho_0) = \sum_{\gamma \in \Gamma} P(\rho \mid \rho_0, \gamma) P(\gamma).$$

$P(\gamma)$ governs the degree to which the expert-defined information influences the prior on ρ , whereas $P(\rho \mid \rho_0, \gamma)$ determines the probability of a candidate partition ρ given ρ_0 and the constraints encoded by γ . The latter builds on the notion of compatibility introduced in [Page et al. \(2022\)](#). Specifically, we say that ρ is compatible with ρ_0 according to γ if ρ_0 can be modified to obtain ρ while satisfying all constraints imposed by γ . Under the specification $P(\rho \mid \rho_0, \gamma) \propto P(\rho) C(\rho \mid \rho_0, \gamma)$, where $C(\rho \mid \rho_0, \gamma)$ is an indicator of such compatibility, this leads to a hard assessment of the expert-defined information. Compatible partitions retain probability proportional to the baseline prior $P(\rho)$, which is independent of ρ_0 , whereas incompatible partitions are assigned zero probability. As a consequence, the prior mass originally assigned to incompatible partitions is redistributed among the compatible ones without otherwise altering their relative weights.

In [Section 1](#), we anticipated that assuming node exchangeability and, consequently, exchangeability of the spectral bands, is not appropriate for our application. Instead, we require the partition ρ to respect the natural ordering of the wavelengths, thereby restricting attention to a smaller subset of all possible partitions. To formalize this idea, we rely on the notion of *admissible partitions* introduced in [Martínez and Mena \(2014\)](#). Specifically, a partition is said to be admissible if and only if for every node $j \in C_r$ and $m \in C_s$ such that $r < s$, then $j < m$. We let $\rho' = (C'_1, \dots, C'_K)$ be an admissible partition, and note that the previous definition requires ordering both the nodes and the clusters and that the following constraints must hold: $1 \in C'_1, p \in C'_K$. Following [Martínez and Mena \(2014\)](#), the collection of all admissible partitions is referred to as the class of set compositions of V , denoted by $\mathcal{C}_V$, whose cardinality equals $|\mathcal{C}_V| = 2^{p-1}$, which is much smaller than the number of all possible partition. The simpler structure of such a space is leveraged to define an equivalent representation of ordered partitions ρ' in terms of ordered allocation vectors $c = (c_1, \dots, c_p)$ which are a convenient reparametrization of allocation vectors z under the assumption of ordered nodes. Specifically, $c_j \in \{0, 1\}$ with $c_j = 1$ if and only if there exists some cluster C'_k such that the j -th node is its largest element. Equivalently,

$$j \in C'_k \iff \sum_{m=1}^{j-1} c_m = k - 1, \quad \text{for } j = 2, \dots, p.$$

By analogy to time-dependent models, where this representation was first introduced, we refer to nodes with $c_j = 1$ as changepoint nodes. By convention, $c_p = 1$, i.e., the final node is always a changepoint node. Following this representation, the number of groups K is easily recovered from c as $K = \sum_{j=1}^p c_j$. Since $\mathcal{C}_V$ is a strict subset of the space of all partitions of p objects, a natural way to define a prior distribution that only places positive mass on admissible partitions ρ' is to assign zero prior probability to all non-admissible partitions while setting a probability proportional to that of a prior on an exchangeable partition with the same cluster sizes $(p_1, \dots, p_K)$. [Pitman \(2006\)](#) shows that the normalizing constant of this operation is straightforward to compute.

We are interested in defining a prior distribution for the random partition that accounts simultaneously for both admissibility with respect to the natural ordering of the wavelength and compatibility with respect to the expert-defined partition. Hence, letting c_0 be the expert-defined ordered allocation vector associated to ρ_0 (which we assume to be ordered) we define $\gamma = (\gamma_1, \dots, \gamma_p)$ as a latent indicator vector such that $\gamma_j = 1$ if the j -th node is forced to

be, or not to be, a changepoint node according to its status in the expert-defined partition. More precisely, $\gamma_j = 1$ guarantees that $c_j = c_{0,j}$ while $\gamma_j = 0$ allows the possibility of c_j to be different from the corresponding quantity in the expert-defined partition. Prior knowledge about the nature of the j -th node can be included in the model by assuming γ to be random with independent components. Namely, $\gamma_j \stackrel{\text{ind}}{\sim} \text{Be}(\eta_j)$ so that values of η_j close to one force the posterior of c_j to be equal to $c_{0,j}$ while small values of η_j indicate that we are unsure about whether the j -th node is a changepoint or not. A crucial hypothesis is that we inform the partition only about the presence of changepoint nodes at specific positions. Namely, we deterministically set $\eta_j = 0$, and consequently $\gamma_j = 0$, for all nodes such that $c_{0,j} = 0$. By doing so, we only allow the possibility of setting $\gamma_j = 1$ when $c_{0,j} = 1$. For coherence, we set $\gamma_p = 1$ since the final node is forced to be a changepoint. The underlying intuition is that we use expert information only to identify locations where a changepoint is believed to occur, while we do not impose prior information on locations where experts indicate the absence of a changepoint. This yields a flexible prior specification in which each expert-defined changepoint may either be retained or disregarded according to the posterior evidence. This choice is also consistent with the application considered here, where expert knowledge is primarily expressed in terms of plausible changepoint locations rather than locations believed not to contain a changepoint. Furthermore, this assumption avoids an over-parametrized prior specification, which is particularly relevant given the hard compatibility constraints imposed by the model. Finally, in Proposition 1, we leverage such an assumption to express $P(\rho \mid \rho_0, \gamma)$ in closed form, without introducing an intractable normalizing constant.

Let $j_1 < \dots < j_H$ denote the ordered indices such that $\gamma_{j_h} = 1$, where $H = \sum_{j=1}^p \gamma_j$. Then, the proposed construction further partitions the nodes into H consecutive subgroups. Setting $j_0 := 0$, the h th subgroup is defined as $\mathcal{I}_h = \{j_{h-1} + 1, \dots, j_h\}$, for $h = 1, \dots, H$. Let $\tilde{p}_h$ denote the number of nodes in the $\mathcal{I}_h$, so that $\sum_{h=1}^H \tilde{p}_h = p$. Then, for any admissible and compatible partition $\rho'_p = (C'_1, \dots, C'_K)$, we define $\rho'_{\tilde{p}_h}$ as the restriction to ρ'_p on $\mathcal{I}_h$ defined as the intersection of any element of ρ'_p with $\mathcal{I}_h$, discarding the empty sets, and preserving their original ordering. Consequently, we write $\rho'_{\tilde{p}_h} = (C_{h,1}, \dots, C_{h,K_h})$, where K_h denotes the number of clusters within subgroup h , and we let $p_{h,k} := |C_{h,k}|$ denote the cardinality of the k th element of $\rho'_{\tilde{p}_h}$, so that $\tilde{p}_h = \sum_{k=1}^{K_h} p_{h,k}$.

PROPOSITION 1. *Let $\rho'_p = (C'_1, \dots, C'_K)$ be an admissible partition and compatible to ρ_0 according to γ . Let $\mathcal{I}_h$, for $h = 1, \dots, H$ be the subgroups determined by conditioning on γ and let $\rho'_{\tilde{p}_h}$ be the restriction to ρ'_p on $\mathcal{I}_h$. Then, $K = \sum_{h=1}^H K_h$ and the following is a proper probability distribution in the space of all admissible and compatible partitions:*

$$(9) \quad \mathbb{P}(\rho = \rho'_p \mid \rho_0, \gamma) = \prod_{h=1}^H \left[\frac{1}{K_h!} \binom{\tilde{p}_h}{p_{h,1}, \dots, p_{h,K_h}} \Pi(p_{h,1}, \dots, p_{h,K_h}) \right],$$

where $\Pi(\cdot)$ can be any Exchangeable Partition Probability Function (EPPF).

The proof is provided in A, while we refer to ??? for a brief overview on exchangeable partition and EPPFs. Equation (9) reduces to the prior on admissible partitions proposed by Martínez and Mena (2014) when $H = 1$, i.e., for $\gamma_j = \mathbf{1}(j = p)$. The latter case is to interpret as when no expert information is defined or enforced and is achieved setting $\eta_j = \mathbf{1}(j = p)$.

As a specific choice for the EPPF, we employ the Poisson-Dirichlet distribution (Perman et al., 1992) that corresponds to the EPPF induced on the partition under a Pitman–Yor process (Pitman and Yor, 1997) prior. Under this choice, Equation (9) simplifies to

$$(10) \quad \mathbb{P}(\rho = \rho'_p \mid \rho_0, \gamma) = \prod_{h=1}^H \frac{\tilde{p}_h!}{K_h!} \frac{\prod_{k=1}^{K_h-1} (\theta + k\sigma)}{(\theta + 1)^{(\tilde{p}_h-1)}} \prod_{k=1}^{K_h} \frac{(1 - \sigma)^{(p_{h,k}-1)}}{p_{h,k}!},$$

where $\sigma \in [0, 1)$ and $\theta > -\sigma$. We refer to Equation (10) as *Informed Ordered Poisson-Dirichlet* prior and we write $\rho \mid \rho_0, \gamma \sim \text{IOPD}(\sigma, \theta; \rho_0, \gamma)$.

Conditionally on γ , the induced prior distribution on the number of groups K_h is preserved with respect to the same quantity computed in the exchangeable case, a result that has been thoroughly studied in the literature (Martínez and Mena, 2014), namely

$$\mathbb{P}(K_h = k \mid \gamma) = \frac{1}{\sigma^k (\theta + 1)_{\tilde{p}_h - 1}} \prod_{i=1}^{k-1} (\theta + i\sigma) (-1)^k C(\tilde{p}_h, k; \sigma),$$

where $C(n, k; \sigma)$ is the central generalized factorial coefficient Charalambides (2002). The prior expected value of the number of groups is given by

$$(11) \quad E(K_h \mid \gamma) = \frac{(\theta + \sigma)_{\tilde{p}_h}}{\sigma (\theta + 1)_{\tilde{p}_h - 1}} - \frac{\theta}{\sigma}.$$

Equation (11) can be used to guide the selection of prior hyperparameters, enabling us to align the expected number of groups with prior beliefs. However, marginalizing the previous results with respect to γ is too involved because of the combinatorial nature of the problem.

The interpretation of the Pitman-Yor hyperparameters requires some care in the proposed informed setting. In the absence of retained expert changepoints, the prior reduces to the usual ordered Pitman-Yor prior on the global partition. In this case, θ plays its standard role as a concentration parameter: larger values of θ generally favor partitions with a larger number of groups. When expert changepoints are retained, however, the interpretation of θ , and subsequently its elicitation, slightly changes because the prior factorizes over the H subgroups induced by γ . In particular, if two expert-defined changepoints $a < b$ are both retained, i.e., $\gamma_a = \gamma_b = 1$, they delimit a subgroup containing the nodes between a and b . Within this subgroup, the quantity $\theta + \sigma$ controls the extent to which the prior favors further partitioning of the nodes, determining the degree of exploration of alternative changepoint configurations between the two retained expert changepoints: larger values of $\theta + \sigma$ lead to a prior propensity for additional groups within the subgroup. **[a me non piace questa parte]**

Wrapping up, the Bayesian hierarchical formulation of the proposed model is

$$(12) \quad \begin{aligned} \mathbf{Y}_i \mid \beta_i, \tau_\varepsilon^2 &\stackrel{\text{ind}}{\sim} \mathbf{N}_L(\Phi \beta_i, \tau_\varepsilon^{-2} \mathbf{I}_L), \quad i = 1, \dots, n \\ \beta_i \mid \mu, \Omega &\stackrel{\text{iid}}{\sim} \mathbf{N}_p(\mu, \Omega^{-1}), \quad i = 1, \dots, n \\ \Omega \mid G &\sim \text{G-Wishart}(d, \mathbf{D}), \\ G \mid \rho &\sim \text{SBM}(\alpha_1, \alpha_2; \rho), \\ \rho \mid \rho_0, \gamma &\sim \text{IOPD}(\sigma, \theta; \rho_0, \gamma). \end{aligned}$$

We complete the model placing the following hyperpriors for the model hyperparameters: $\tau_\varepsilon^2 \sim \text{Inv-Gamma}(a_\varepsilon, b_\varepsilon)$, $\mu \sim \mathbf{N}_p(\mathbf{0}, \tau_\mu^{-2} \mathbf{I}_p)$, $\theta \mid \sigma \sim \text{Gamma}_{(-\sigma)}(a_\theta, b_\theta)$, where $\text{Gamma}_{(-\sigma)}$ is $(-\sigma)$ -shifted Gamma distributed with parameters (a_θ, b_θ) , i.e., $\theta + \sigma$ is gamma distributed with parameters (a_θ, b_θ) (Martínez and Mena, 2014), $\sigma \sim \text{Beta}(a_\sigma, b_\sigma)$, while $\gamma_j \stackrel{\text{ind}}{\sim} \text{Bernoulli}(\eta)$, for each $j \in \{1, \dots, p-1\}$ such that $c_{0,j} = 1$ while we deterministically assign $\gamma_j = 0$ otherwise. We recall that $\gamma_p = 1$ by construction.

5. Posterior sampling strategy. We perform posterior inference using a blocked Gibbs sampling strategy. Let $\Delta = (\tau_\varepsilon^2, \beta, \mu, G, \Omega, \gamma, \rho, \sigma, \theta)$ denote the collection of model parameters and latent variables. We partition Δ into three blocks, $\Delta_1 = (\tau_\varepsilon^2, \beta, \mu)$, $\Delta_2 = (G, \Omega)$, and $\Delta_3 = (\gamma, \rho, \sigma, \theta)$. The sampling scheme proceeds by iteratively updating each block,

conditionally on all the others. We first update Δ_1 by sampling τ_ε^2 , the spline coefficients β , and the mean vector μ from their full conditional distributions, following Codazzi et al. (2022). Next, we update Δ_2 by first sampling G through a reversible-jump MCMC scheme (Dobra et al., 2011), slightly modified to account for the partition-informed SBM prior on the graph, and then by sampling $\Omega \mid G$ from its G -Wishart full conditional distribution using the sampling method of Lenkoski and Dobra (2011). Finally, we update the partition-related block Δ_3 . To do so, we first update the latent indicators γ component-wise, conditionally on the ordered partition ρ'_p , by sampling each γ_j from its full conditional distribution. The resulting configuration of γ determines the H subgroups used in the subsequent partition update. **Then, we update the partition ρ using Metropolis-Hastings moves within each of the subgroups.** Specifically, we employ split/merge moves to modify the number of clusters and shuffle moves to redistribute nodes between adjacent clusters, keeping their number fixed. Candidate change-point locations are selected according to adaptive proposal weights following the approach of Benson and Friel (2018). After all subgroup-specific updates have been performed, the resulting local partitions are combined to obtain the updated global partition ρ . The step is concluded by updating the hyperparameters σ and θ , whose full conditional distributions are given in Martínez and Mena (2014). A detailed description of all full conditional distributions and Metropolis-Hastings updates is provided in Appendix B. The code is publicly available at ???.

6. Simulation study.

6.1. Data generation and model comparison. In this simulation study, we assess the ability of the proposed model to recover the underlying graph, the node partition and the precision matrix. To this end, we set a true graph G_{true} with $p = 40$ nodes and a block structure based on the node partition ρ_{true} with $K_{\text{true}} = 8$ clusters, see Figure 2. We generate $B = 50$ independent datasets from the Gaussian graphical model in (1)-(2) for each sample sizes $n \in \{200, 500\}$. For each dataset, the precision matrix Ω is assumed Markov with respect to G_{true} and drawn from a G-Wishart distribution with shape parameter $d = 10$ and $\mathbf{D}$ equal to the identity matrix.

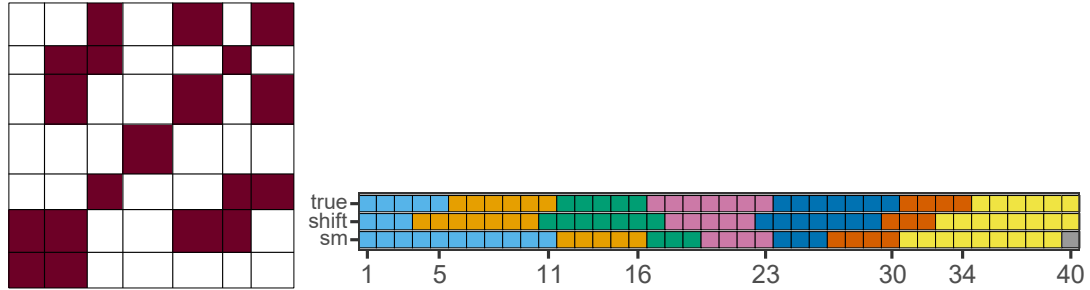


Fig 2: Left panel: adjacency matrix of the true graph. Right panel: expert-based partitions used in the three simulation scenarios, i.e., the true, shifted, and split/merge partitions. **[sul grafo metterei i label con gli stessi valori presenti nel grafico a destra]**

In order to assess the robustness of the model with respect to the expert information, we consider three different simulation scenarios according to different expert partitions. In the first scenario, the expert is assumed to be an oracle, so $\rho_0 = \rho_{\text{true}}$. The remaining two scenarios investigate the effect of misspecified expert information, i.e., $\rho_0 = \rho_{\text{shift}}$, and $\rho_0 = \rho_{\text{sm}}$. The partition ρ_{shift} is obtained by slightly shifting the true changepoint locations, mimicking

experts who identify approximately correct block boundaries. In contrast, ρ_{sm} is constructed by splitting and merging neighboring blocks, representing a form of misspecification in which some changepoints are omitted while others are spuriously introduced. See Figure 2 for a visual representation of the expert-based partitions over the three scenarios.

In each simulation scenario, we consider different values of the parameter η which controls the weight assigned to the prior information provided by the experts. Specifically, we consider $\eta \in \{0, 0.5, 0.75, 0.9\}$, where increasing values of η give progressively more weight to the expert partition, while $\eta = 0$ corresponds to the uninformed random partition model. As a benchmark, we also compare our approach with the model of Codazzi et al. (2022) which does not consider any partition of the nodes. In the following, the latter is referred to as No-Partition (NP) model.

To obtain a posterior point estimate of the partition, we employ the Binder loss criterion (Binder, 1978), adapted to ordered partitions as described by Corradin et al. (2022). As an alternative, we can estimate the partition by identifying as changepoints the nodes whose posterior probabilities exceed a specific threshold. For instance, with a threshold of 0.5, the resulting estimates are similar to the ones obtained using the Binder loss criterion. The graph is estimated by thresholding the posterior edge-inclusion probabilities. Following Codazzi et al. (2022) and Colombi et al. (2024), the threshold is chosen to control the Bayesian false discovery rate (BFDR; Müller et al. 2007).

The accuracy in recovering the true partition is evaluated through the Adjusted Rand Index (ARI; Hubert and Arabie 1985), computed with respect to the true partition; the index ranges between zero (the two clustering do not agree on any pair of points) and one (the clustering are the same). To assess graph recovery, we follow Mohammadi and Wit (2015) and Colombi et al. (2024) considering two complementary measures. The first is the Standardized Structural Hamming Distance (Std-SHD; Tsamardinos et al. 2006), which, for undirected graphs, corresponds to the proportion of incorrectly estimated edges. The second is the F1-score [citazione], which, unlike the Std-SHD, does not equally weight missing (including) edges that are present (absent) in the true graph. These two quantities are defined as

$$(13) \quad \text{Std-SHD} = (FP + FN) / \binom{p}{2}, \quad \text{F1-score} = (2TP) / (2TP + FP + FN),$$

where FP, FN and TP denote the numbers of false positives, false negatives, and true positives, respectively. ~~Finally, precision matrix recovery is assessed through the 2-Wasserstein distance between the zero-mean Gaussian distributions with covariance matrices $\Sigma_{\text{true}} = \Omega_{\text{true}}^{-1}$ and $\Sigma_{\text{est}} = \Omega_{\text{est}}^{-1}$. Lower values of Std-SHD and 2-Wasserstein, as well as higher values of ARI and F1-score, indicate better recovery.~~

6.2. Results. [luci da fare: accorpare i tre scenari e mettere le figure organizzate per metrica, con $n = 500$ in appendix]

6.3. *Scenario 1: $\rho_0 = \rho_{\text{true}}$.* The following experts-as-oracle setting highlights the beneficial effect of centering the prior increasingly close to the available prior information through the choice of the parameter η . Indeed, under the uninformed partition model ($\eta = 0$), the recovery of the underlying partition is substantially poorer than in the corresponding informed cases, both for $n = 200$ and $n = 500$, (left panel of Figure 3). Moreover, increasing $\eta = 0.5$ to either $\eta = 0.75$ or $\eta = 0.9$ leads to further improvements when $n = 200$, as the variability is considerably reduced. Eventually, for $n = 500$, the ARI is very close to one for all positive values of η , as the information from the data overrides the one from the prior.

The advantages of incorporating partition information are also reflected in graph recovery. The NP model exhibits the largest Std-SHD (right panel of Figure 3), whereas the partition-based models achieve substantially smaller values. Moreover, the Std-SHD generally decreases

as η increases, suggesting that stronger confidence in a correctly specified expert partition translates into improved graph estimation. A similar pattern is observed for the F1-score (left panel of Figure 9 in Appendix).

~~The same qualitative behaviour is observed for precision matrix recovery. The model without partitioning produces the largest 2-Wasserstein distances (right panel of Figure 9 in Appendix), whereas all partition-based models yield substantially smaller values.~~

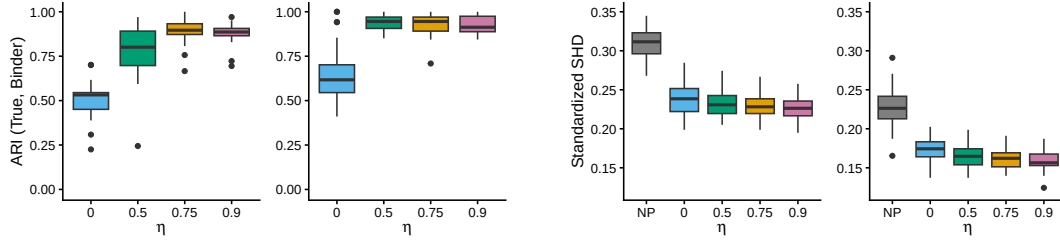


Fig 3: **[aggiustare caption]** $\rho_0 = \rho_{\text{true}}$ setting. For each quantity, the left and right sub-panels correspond to $n = 200$ and $n = 500$, respectively.

6.4. Scenario 2: for $\rho_0 = \rho_{\text{shift}}$. In this setting, the expert partition is obtained by shifting the true changepoint locations. It follows that the estimated ARI (left panel of Figure 4) values for $\eta > 0$ do not converge to one as fast as in the experts-as-oracles setting but settle at levels that are more comparable to those of the uninformed partition model ($\eta = 0$) case. However, informed partition models achieve higher ARI values than the uninformed, especially as n increases, suggesting that even an approximately correct expert partition provides useful prior information and that, as the sample size increases, our model is better able to filter out the amount of correct information contained in ρ_0 . We highlight that, as η increases, the estimated partition becomes increasingly aligned with the experts' partition, but the ARI (Figure 12a in the Appendix) remains stable, indicating that incorporating increasing amounts of expert information does not impair partition recovery, even when the experts' partition is inconsistent with ρ_0 .

As in the previous scenario, the model without partitioning exhibits the largest Std-SHD (right panel of Figure 4), whereas all partition-based models substantially improve graph estimation. The effect of η is less pronounced than when the expert partition is correctly specified. Although the Std-SHD shows a slight decreasing trend as η increases, the differences among the informed models are relatively small. The F1-score (left panel of Figure 10) exhibits the same behaviour.

~~Precision matrix recovery displays essentially the same behaviour as in the correctly specified scenario (right panel of Figure 10).~~

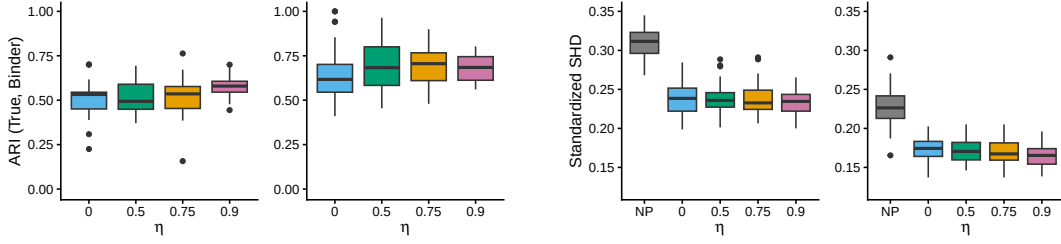


Fig 4: $\rho_0 = \rho_{\text{shift}}$ setting. For each quantity, the left and right sub-panels correspond to $n = 200$ and $n = 500$, respectively.

6.5. *Scenario 3: $\rho_0 = \rho_{sm}$.* In this setting the expert partition is specified through a split-and-merge configuration. Here, the proposed informed partition model continues to exhibit a remarkable degree of robustness. By inspecting the ARI (left panel of Figure 5), we see that the informed models achieve a better partition recovery than the uninformed one. Consistently with the previous setting, as η increases the estimated partition aligns with the partition provided by the experts (Figure 12b in the Appendix) while remaining below one even for the largest values of η , suggesting again that the proposed model is able to depart from an incorrect expert specification whenever supported by the data.

The graph recovery results closely resemble those obtained under the shifted partition specification. The model without partitioning again exhibits the largest Std-SHD (right panel of Figure 5), while all partition-based models provide a substantially more accurate reconstruction of the underlying graph. Increasing η yields modest improvements in Std-SHD, and the same behaviour is observed for the F1-score (left panel of Figure 11).

~~The results for precision-matrix recovery are consistent with the previous scenarios (right panel of Figure 5).~~

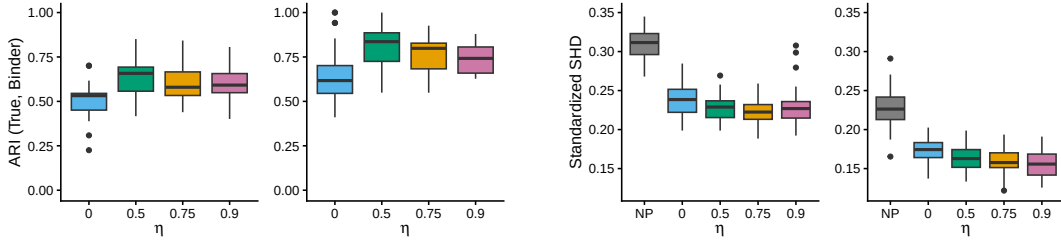


Fig 5: $\rho_0 = \rho_{sm}$ setting. For each quantity, the left and right sub-panels correspond to $n = 200$ and $n = 500$, respectively.

Overall, the simulation study highlights three main findings. First, incorporating partition information substantially improves the recovery of the latent partition, graph, and precision matrix compared with the model without partitioning. Second, the proposed informed partition model effectively exploits accurate expert information and remains robust to moderate mis-specification of the expert partition, as evidenced by the results obtained under both the shifted and split-and-merge scenarios. Finally, the comparison between the two sample sizes indicates that expert information appears to be particularly beneficial when the available sample size is relatively small, whereas its contribution naturally decreases as more observations become available and the likelihood becomes increasingly informative.

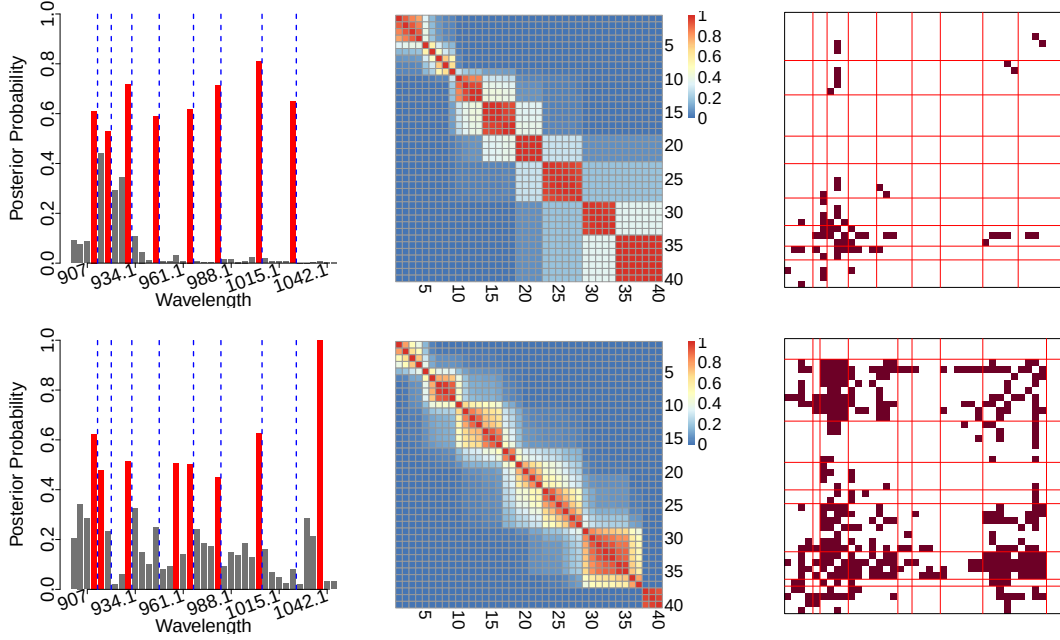


Fig 6: Posterior quantities for strawberry (top row) and non-strawberry (bottom row) purees for $\eta = 0.75$. For the changepoint plots, red bars denote the estimated partition, while blue dotted lines indicate the partition informed by experts. For the graph plot, red bars denote the estimated partition.

7. Analysis of fruit purees data.

7.1. Posterior inference for $\eta = 0.75$. We analyse the strawberry and non-strawberry purees dataset described in Section 2. The dataset comprises $n_{ns} = 632$ non-strawberry and $n_s = 351$ strawberry purees curves measured on an equally spaced grid of 235 wavelengths over the interval $I = [899.327, 1802.564] \text{ nm}^{-1}$. As in Codazzi et al. (2022), we represent the observed spectra using $p = 40$ B-spline basis functions, a choice motivated by the trade-off between accurately capturing the shape of the smoothed curves and limiting the computational burden. The resulting p latent spline coefficients provide a lower-dimensional representation of the spectra, on which we perform the subsequent graphical and partition analyses. Thus, the nodes of the graph and the partition correspond to the latent spline coefficients, and indirectly to the spectral regions represented by the corresponding basis functions. **[non mi fa impazzire come incipit ma per ora ok][sono d'accordo, poi lo miglioriamo]**

We perform our analysis for $\eta \in \{0, 0.5, 0.75, 0.9\}$, covering a range from null to strong prior belief. Moreover, to assess the contribution of the latent node partition in estimating the graph, we also compare our model with Codazzi et al. (2022), which does not consider any partition of the nodes. We report the results for $\eta = 0.75$ here and defer the results for the remaining values of η to Appendix E. We set the SBM parameters α_1 and α_2 to be equal to 0.75, which implies $\mathbb{E}(Q_{rs}) = 0.5$ and $\text{Var}(Q_{rs}) = 0.1$, favouring values close to 0 or 1. **[quali valori??]** For what concerns partition hyperparameters, we place a uniform prior on σ , while the specification of θ differs for $\eta = 0$ and $\eta > 0$. In the first case, we fix the prior expected number of cluster, and its variance, equal to $\mathbb{E}[K] = 10$ and $\text{Var}(K) = 1$, respectively. **[controllare che K sia stato definito]** Such configuration is achieved by setting $a_\theta = 0.90$ and $b_\theta = 0.95$. When $\eta > 0$, we set $a_\theta = 0.88$ and $b_\theta = 0.94$, which corresponds to $\mathbb{E}[K_h] = 4$ and $\text{Var}(K_h) = 1$ for $h = 1, \dots, H$, considering an heuristic assumption of

having 5 subgroups of average size around 8. The remaining hyperparameters are set to $d = 3$, $D = \mathbf{I}p$, $a_\varepsilon = 10$, $b_\varepsilon = 0.001$, $\tau_\mu^{-2} = 100$.

The Gibbs sampler was run for a total of 200,000 iterations, with the first 50,000 discarded as burn-in, and a thinning of 10. The total computational time, computed on a Windows 11 laptop equipped with an Intel Core i5-1035G1 processor and 8 GB of RAM, was approximately 30 minutes for $\eta = 0$, and between 40 and 80 minutes for $\eta \in \{0.5, 0.75, 0.9\}$. The computational time generally increases with η , as stronger prior information leads to a more computationally demanding partition update, and is higher for the non-strawberry purees due to their larger sample size.

To illustrate the main features of posterior inference, we first consider the random partition. Figure 6 (top and bottom left panels) reports the posterior probabilities of a changepoint together with the estimated partition. Red bars denote the estimated partition, whereas blue dotted lines indicate the partition elicited by the experts. The resulting point estimates contain nine groups for both the strawberry and non-strawberry curves. Overall, they are broadly consistent with the nine-group partition elicited by the experts, while also exhibiting some interesting differences. Importantly, the model does not simply reproduce the expert-defined partition. For the non-strawberry curves, three changepoints identified by the experts receive little or no posterior support, while three alternative changepoints, not anticipated by the experts, are identified with high posterior probability. For the strawberry curves, instead, the agreement with the expert-defined partition is substantially closer, with no deviations in the resulting point estimate but with some uncertainty in the first part of the wavelength. A further feature of the point estimate is that, for the non-strawberry curves, both nodes 4 and 5 are identified as changepoints, whereas the experts identified only node 4. This may indicate a region in which the precise location of the changepoint is not sharply determined by the posterior distribution. In particular, when adjacent nodes receive substantial posterior support as potential changepoints, the minimization of Binder’s loss over ordered partitions can result in both nodes being selected in the point estimate. One possible refinement would be to retain only the adjacent node with the highest posterior changepoint probability. We do not introduce such a post-processing step here, but note it as a possible extension of the point-estimation procedure. Figure 7 provides a visual comparison of the two estimated partitions. The partitions agree on the majority of changepoint nodes (4, 9, 18, 22, 28), while differences arise at several other locations. The corresponding ARI is 0.648, indicating agreement between the two estimated partitions, but reflecting also some differences in their allocation of changepoint nodes.

The top and bottom center panels of Figure 6 report the posterior coclustering probabilities for the strawberry and non-strawberry curves, respectively. The strawberry case exhibits a comparatively well-defined block structure, with only the earlier part of the spectral band presenting some uncertainty with coclustering probability around 0.25 to 0.45. In contrast, the non-strawberry case shows substantially greater uncertainty, with more intermediate coclustering probabilities.

Finally, the estimated graphs together with their corresponding estimated partitions, shown in the top and bottom right panels of Figure 6, reveal markedly different dependence structures for the two types of curves. The graph for the strawberry curves is relatively sparse, with a density of 0.12, whereas the non-strawberry graph is substantially denser, with approximately 0.58 of the possible edges present. This difference is consistent with the stronger conditional dependence structure observed for the non-strawberry curves. The partitions overlaid on the graphs also reveal distinct patterns in the organization of this dependence. For the strawberry curves, the strongest connectivity is concentrated among the first groups, with very limited connectivity involving the final groups. In contrast, the non-strawberry graph exhibits several clearly defined blocks, characterized by substantial within-group connectivity in the first

part of the spectrum, together with interesting connections between groups. In particular, pronounced connections are observed between the second-third groups and the fourth group, as well as between the second-third groups and the penultimate group. The latter group also exhibits notable connections with the first, fourth, and fifth groups.

Additional results, including a comparison with different η , are reported in Appendix E.

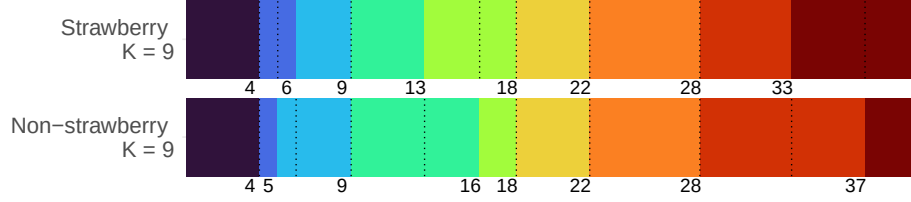


Fig 7: Estimated partition comparison between strawberry and non-strawberry spectra for $\eta = 0.75$.

7.2. Distance between strawberry and non-strawberry distribution of coefficients. In this section, we argue that the main difference between strawberry and non-strawberry spectra lies in the underlying conditional dependence structure of the chemical groups forming the compound, and not in the shape of the spectra themselves.

This claim is first supported by a visual comparison of the estimated basis expansion coefficients (see Figure 8), which exhibit very similar patterns between the two groups. In contrast, the estimated graphs (for example the graph estimated for $\eta = 0.75$, see Figure 6) reveal substantial differences between the corresponding conditional dependence. Since, under our model, structural differences in the graph translate into differences in the distribution of the latent coefficients, the graphical component appears to be the main source of discrepancy between strawberry and non-strawberry purees.

We further support this claim quantitatively by measuring the distance between the posterior distributions of the basis expansion coefficients estimated under the two groups. To this end, we use the squared 2-Wasserstein distance. Recall that, when comparing two multivariate normal distributions with means μ_s and μ_{ns} and covariance matrices Σ_s and Σ_{ns} , respectively, this distance admits the closed-form expression:

$$(14) \quad W_2^2(s, ns) = \|\mu_s - \mu_{ns}\|^2 + \text{tr} \left(\Sigma_s + \Sigma_{ns} - 2(\Sigma_s^{1/2} \Sigma_{ns} \Sigma_s^{1/2})^{1/2} \right).$$

The expression in (14) naturally separates the discrepancy between the two distributions into a component due to differences in their means and a component due to differences in their covariance structures. This decomposition allows us to assess whether the observed separation between strawberry and non-strawberry curves is primarily associated with their mean coefficient profiles or with their dependence structure. Across all considered values of η , the covariance component accounts for more than 90% of the total squared Wasserstein distance, with values ranging from 90.7% to 91.1% and negligible variation across prior specifications. This provides quantitative evidence that the differences between the two groups are predominantly driven by their covariance structure rather than by differences in the mean coefficient profiles. Further details are provided in Appendix D.

7.3. Classification of new curves. [accorciare molto il seguente rimuovendo alcune cose e rivedere la notazione]

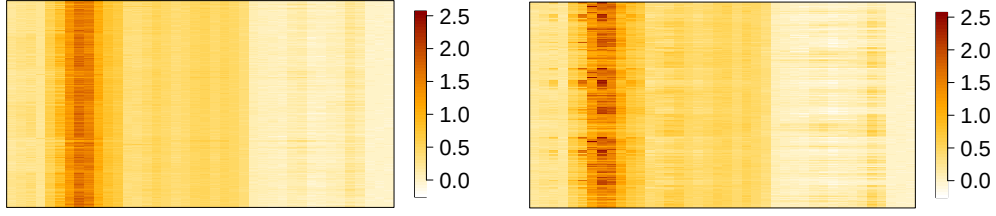


Fig 8: Posterior β coefficients' values for strawberry (left) and non-strawberry (right) dataset. Rows represent observations, columns the $p = 40$ elements

We frame the classification problem of distinguish between strawberry and non-strawberry spectra within a Bayesian decision-theoretic framework (Gelman et al., 2013, Chapter 9). Intuitively, we assign a label $d \in \mathcal{L} := \{s, ns\}$ to any new curve $\tilde{Y}$ so that it maximizes a suitably defined utility function that measure how similar $\tilde{Y}$ is to the observations in class d , while integrating over the posterior uncertainty. Specifically, the outcome of the decision that assigns label d to the new curve is the latent smoothed representation of $\tilde{Y}$, which, in our model, is represented by its basis expansion coefficient $\tilde{\beta}$ alone, since the B-spline bases is common to both classes. However, the latter is not directly observable but it is a random quantity that depends on the decision d (which is not random), the training dataset $\mathbf{Y}_d$ as well as the new curve $\tilde{Y}$. Its law can be expressed as the following predictive distribution:

$$(15) \quad P(\tilde{\beta} | d, \mathbf{Y}_d, \tilde{Y}) = \int P(\tilde{\beta} | d, \mathbf{Y}_d, \tilde{Y}, \Theta_d) P(\Theta_d | d, \mathbf{Y}_d, \tilde{Y}) d\Theta_d,$$

where $\Theta_d = (\mu_d, \Omega_d, \tau_{\varepsilon,d}^2)$ and the integration occurs with respect to the posterior distribution of the parameters in class d . The conditional distribution in Equation (15) is $P(\tilde{\beta} | d, \mathbf{Y}_d, \tilde{Y}, \Theta_d) = \mathcal{N}_p(\mathbf{b}_d, \mathbf{B}_d^{-1})$, whose parameter are and $\mathbf{B}_d = \tau_{\varepsilon,d}^2 \Phi^\top \Phi + \Omega_d$, $\mathbf{b}_d = \mathbf{B}_d^{-1} [\tau_{\varepsilon,d}^2 \Phi^\top \tilde{Y} + \Omega_d \mu_d]$. We define the utility, or reward, associated to decision d and outcome $\tilde{\beta}$ as

$$U_d(\tilde{\beta}) = \sum_{i=1}^{n_d} \alpha_{d,i} \kappa_d(\tilde{\beta}, \beta_{d,i}),$$

where n_d denotes the number of training curves belonging to class d , and

$$\kappa_d(\tilde{\beta}, \beta_{d,i}) = \exp \left\{ -(\tilde{\beta} - \beta_{d,i})^\top \Omega_d (\tilde{\beta} - \beta_{d,i}) \right\}.$$

Intuitively, the similarity between two coefficient vectors is measured through a Gaussian kernel with class-specific precision matrix Ω_d . The latter extends Zheng et al. (2019), who relied on an isotropic Gaussian kernel. The utility weights are then defined by

$$\alpha_d = (\kappa_d(\beta_{d,i}, \beta_{d,j}) + C^{-1} \mathbf{I}_{n_d})^{-1} \mathbf{1}_{n_d},$$

where $C > 0$ is a regularization parameter and $\mathbf{1}_{n_d}$ denotes the n_d -dimensional vector of ones.

Finally, we label the new spectrum $\tilde{Y}$ as a strawberry if

$$(16) \quad P(U_s(\tilde{\beta}) > U_{ns}(\tilde{\beta}) | \mathbf{Y}_s, \mathbf{Y}_{ns}, \tilde{Y}),$$

and we label it as an non-strawberry spectrum otherwise.

Practically, we assessed classification performance by randomly selecting 150 observations from the strawberry class and 150 observations from the non-strawberry class to form the

test set, for a total of 300 test curves. The test set represents approximately 30% of the available observations. The remaining curves were used for model fitting. For each test curve $\tilde{Y}$, we generated posterior samples of the latent coefficient vector by sampling from the conditional distribution in (15) at each MCMC draw of the class-specific parameters Θ_d . At each MCMC iteration t , the sampled vector $\tilde{\beta}^{(t)}$ was used to evaluate both class-specific utilities, $U_s^{(t)} = U_s(\tilde{\beta}^{(t)})$ and $U_{ns}^{(t)} = U_{ns}(\tilde{\beta}^{(t)})$. The posterior probability that the strawberry class yields a larger utility is then approximated by the Monte Carlo frequency

$$\hat{p}_s = \frac{1}{T} \sum_{t=1}^T \mathbb{I} \left\{ U_s^{(t)} > U_{ns}^{(t)} \right\},$$

which provides a Monte Carlo approximation to (16). The uncertainty in $\tilde{\beta}$ is therefore propagated directly into the classification probability. The final classification is performed by comparing $\hat{p}_s$ with a threshold $\zeta \in (0, 1)$. The resulting decision rule is $d^* = s$ if $\hat{p}_s > \zeta$, $d^* = ns$ otherwise. The classification threshold is selected through classical ROC analysis by maximizing the Youden index J (Youden, 1950; Fluss et al., 2005; Le, 2006) on the test set.

We evaluated the classification performance under the three levels of expert prior information considered in the main manuscript, $\eta \in \{0.5, 0.75, 0.9\}$, and under a baseline specification in which no partition information was incorporated into the prior on the graph underlying the precision matrix (Codazzi et al., 2022). The results were highly consistent across these specifications, with the lower accuracy achieved among all the cases being of 96.3% and the higher being of 96.7%. In all cases, only one strawberry puree is misclassified, resulting in a sensitivity of 99.3% for the strawberry class. The complete set of confusion matrices for all values of η and for the baseline specification is reported in Tables 4-7 in Appendix F.

Our predictive performance is in line with that reported in the work of Zhang et al. (2019) which focuses only on the predictive task. These results are particularly encouraging in light of the potential trade-off between inferential interpretability and predictive performance. In our application, incorporating expert information into the prior does not compromise classification accuracy while providing additional information that facilitates recovery and interpretation of the latent dependence structure underlying the functional observations.

8. Discussion. [commenti:] un limite è che possiamo solo aumentare i cluster all'aumentare della prior strength negli esperti.

estendere la G - Wishart con la discrete spike and slab di Sulem.

dal punto di vista applicato, ci serve un modello vero di classificazione che non sia un semplice by product perché noi in primis vogliamo fare inferenza, e byproduct è classificazione. Se uno vuole fare adulteration detection vera e propria: joint model first, e qualcosa di più specifico

In our application, the proposed IOPD prior for the partition in conjunction to the SBM prior for the graph illustrate the two complementary roles of expert information in Bayesian graph learning. On the one hand, it allows expert knowledge to exert a substantial influence on the inferred partition, with the strength of this influence controlled by η . On the other hand, it retains sufficient flexibility for the observed data to support deviations from the expert-defined structure. This balance is particularly relevant in our spectroscopy application, where domain knowledge can provide valuable information about meaningful spectral regions, but the dependence structure among wavelengths may also reveal patterns that are not fully captured by prior knowledge. The resulting partition provides a meaningful organization of the conditional dependence structure, grouping spectral regions that exhibit similar patterns of connectivity within the estimated graph. This offers an interpretable way to identify groups of spectral coefficients whose conditional relationships are jointly structured, and so to discover

dependence patterns over the wavelength. In our application, this perspective highlights a clear difference between strawberry and non-strawberry curves. The non-strawberry curves exhibit a substantially denser network and a more pronounced block structure, with several groups showing strong within-group connectivity as well as connections across groups. In contrast, the strawberry curves yield a sparser dependence structure, with connectivity more concentrated in a limited portion of the spectrum. These differences suggest that the combination of expert-informed partitioning and graphical modeling can provide a useful representation of class-specific dependence patterns in spectroscopy, potentially offering additional information for understanding the spectral structure associated with food adulteration.

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APPENDIX A: PROOF OF PROPOSITION 1

Consider, without loss of generality, $H = 2$, and let $j_1 = j$ and $j_2 = p$, for some positive integer $j < p$. Consequently, $\mathcal{I}_1 = \{1, \dots, j\}$ and $\mathcal{I}_2 = \{j+1, \dots, p\}$. From the compatibility constraints imposed by γ on ρ'_p , it follows that $c_j = c_{0,j}$. Furthermore, under the assumption that only changepoint nodes in the expert-defined partition may have positive values of γ , it follows that $c_j = 1$. Then, let $k_j \in \{1, \dots, K\}$ be such that $j \in C'_{k_j}$. Let $m \in \{1, \dots, p\}$, $m \neq j$, be any node of the graph different from j . Then,

- (i) if $m < j$, then $m \in C'_k$ for some positive integer $k \leq k_j$. This follows from the ordering constraints imposed by the admissibility of ρ'_p .
- (ii) if $m > j$, then $m \in C'_k$ for some positive integer $k > k_j$. The inequality $k \geq k_j$ follows from the ordering constraints imposed by the admissibility of ρ'_p , while the strict inequality follows from the definition of $c_j = 1$.

It then follows that there cannot exist any cluster C'_k such that $C'_k \cap \mathcal{I}_1 \neq \emptyset$ and $C'_k \cap \mathcal{I}_2 \neq \emptyset$. Indeed, by admissibility, any such cluster C'_k would contain at least $\{j, j+1\}$, which is not possible because of (ii). Consequently, $\rho'_{\tilde{p}_1} = (C_{1,1}, \dots, C_{1,K_1})$ and $\rho'_{\tilde{p}_2} = (C_{2,1}, \dots, C_{2,K_2})$ are disjoint, and $\mathcal{I}_1 \cup \mathcal{I}_2 = \{1, \dots, p\}$ implies that $K_1 + K_2 = K$.

Fix $h \in \{1, 2\}$. By construction, $\rho'_{\tilde{p}_h}$ is an admissible partition of $\mathcal{I}_h$, since ρ'_p is an admissible partition of $\{1, \dots, p\}$. From [Martínez and Mena \(2014\)](#), the following is a proper probability distribution on the set of compositions of $\mathcal{I}_h$:

$$P(\rho'_{\tilde{p}_h}) = \frac{1}{K_h!} \binom{\tilde{p}_h}{p_{h,1}, \dots, p_{h,K_h}} \Pi(p_{h,1}, \dots, p_{h,K_h}),$$

where $\Pi(\cdot)$ can be any EPPF and $p_{h,k} = |C_{h,k}|$. Finally, the statement follows because we have shown that the compatibility assumption ensures that the space of admissible partitions compatible with γ is the product space of the sets of compositions of $\mathcal{I}_1$ and $\mathcal{I}_2$, since any cluster C'_k must be contained either in $\mathcal{I}_1$ or in $\mathcal{I}_2$.

APPENDIX B: DETAILED MCMC SCHEME

B.1. Pseudocode.

Algorithm 1: Gibbs sampler for the proposed model.

For each iteration:

- Step 1.** Update the functional parameters $(\tau_\varepsilon^2, \beta, \mu)$ from their corresponding full conditional distributions (Equations (17), (18), (19)).
 - Step 2.** Update (G, Ω) conditional on the current partition ρ' and the centered spline coefficients using the reversible-jump MCMC scheme described in Section B.4.
 - Step 3.** Update the latent indicators γ conditional on the current partition ρ' as described in Section B.5.
 - Step 4.** Update the ordered partition ρ' conditional on (G, γ) as described in Section B.6.
 - Step 5.** Update σ and θ from their full conditional distributions as described in Section B.7.
-

B.2. Update of the data variance. For the variance of the data $\mathbf{Y}_i$, $i = 1, \dots, n$, τ_ε^{-2} , we sample from

$$(17) \quad \tau_\varepsilon^{-2} \mid \text{rest} \sim \text{IG} \left(a + \frac{nr}{2}, \frac{1}{2} \left[2b + \sum_{i=1}^n (\mathbf{Y}_i - \Phi \beta_i)^\top (\mathbf{Y}_i - \Phi \beta_i) \right] \right).$$

B.3. Update of the Gaussian Graphical Model Components. We sample the β_i coefficients from their full conditional distribution given by

$$(18) \quad \beta_i \mid \text{rest} \sim \mathcal{N}_p(\mathbf{b}_i, \mathbf{B}),$$

where $\mathbf{B} = \left(\frac{\Phi^\top \Phi}{\tau_\varepsilon^2} + \Omega \right)^{-1}$ and $\mathbf{b}_i = \mathbf{B} \left(\frac{\Phi^\top \mathbf{Y}_i}{\tau_\varepsilon^2} + \Omega \boldsymbol{\mu} \right)$, $i = 1, \dots, n$.

The full conditional distribution of $\boldsymbol{\mu}$ is given by

$$(19) \quad \boldsymbol{\mu} \mid \text{rest} \sim \mathcal{N}_p(\mathbf{m}, \mathbf{M}),$$

where $\mathbf{M} = \left(\frac{\mathbf{I}_p}{\sigma_\mu^2} + n \Omega \right)^{-1}$ and $\mathbf{m} = \mathbf{M} \Omega \sum_{i=1}^n \beta_i$.

B.4. Update of the Precision Matrix and the Graph. Conditional on the current ordered partition ρ'_p and the corresponding group allocation $\mathbf{z} = (z_1, \dots, z_p)$, we update the graph G and the precision matrix Ω using the current centered spline coefficients $\tilde{\beta}_1, \dots, \tilde{\beta}_n$, where $\tilde{\beta}_i = \beta_i - \boldsymbol{\mu}$. Let $\tilde{\mathbf{B}} = (\tilde{\beta}_1, \dots, \tilde{\beta}_n)^\top$ and define $\mathbf{S} = \tilde{\mathbf{B}}^\top \tilde{\mathbf{B}}$. We use the G -Wishart prior for the precision matrix Ω in Equation (3), so that, conditional on G and the data,

$$(20) \quad \Omega \mid G, \tilde{\mathbf{B}} \sim \text{G-Wishart}(d + n, \mathbf{D} + \mathbf{S}).$$

We update G using a reversible-jump MCMC scheme (Mohammadi and Wit, 2019) based on single-edge birth and death moves. At each iteration, we uniformly select one edge (i, j) and propose to toggle its presence in the graph. The acceptance probability combines the contribution of the G -Wishart model with the partition-informed prior on the graph.

To describe the latter contribution, let p_s denote the cardinality of group s induced by ρ'_p , and let

$$S_{uv}(G) = \#\{(i, j) \in E(G) : z_i = u, z_j = v\}$$

denote the number of currently active edges between groups u and v . The total number of possible edges between the two groups is

$$T_{uv} = \begin{cases} p_u p_v, & u \neq v, \\ \binom{p_u}{2}, & u = v. \end{cases}$$

Recall that, under the SBM prior, edges between groups u and v are conditionally independent Bernoulli variables with common inclusion probability Q_{uv} , where $Q_{uv} \sim \text{Beta}(\alpha_1, \alpha_2)$. Integrating out Q_{uv} yields Equation (7). Consequently, adding/deleting an edge contributes to the acceptance ratio via

$$\frac{S_{uv} + \alpha_1}{T_{uv} - S_{uv} - 1 + \alpha_2}, \quad \text{or} \quad \frac{T_{uv} - S_{uv} + \alpha_2}{S_{uv} - 1 + \alpha_1},$$

the former in the case of proposing an edge addition between groups u and v , and the latter in the case of proposing a deletion.

Thus, conditional on the current partition, the graph update depends on the numbers of active and inactive edges within and between the induced groups. Together with the contribution of the G -Wishart model, these terms determine the acceptance probability for the proposed graph move. For more details we refer to Mohammadi and Wit (2019). **[dico "for more details..etc" perché non ho inserito tutto nell acceptance ratio. Va bene o lasciamo così o devo inserire anche i termini dalla G wishart?]** After the graph update, we draw the precision matrix conditional on the resulting graph according to (20) using the G -Wishart sampler described in Algorithm 1 of Mohammadi and Wit (2019). The resulting graph and precision matrix constitute a draw from the graph-precision update conditional on the current partition.

B.5. Update of the Auxiliary Variables. [la notazione è molto pesante... Forse è semplificabile, ne dovremmo parlare insieme]

We update the auxiliary vector $\gamma = (\gamma_1, \dots, \gamma_p)^\top$ component-wise within each MCMC iteration. Recall that γ_j determines whether the changepoint status of node j is enforced according to the expert-defined ordered partition. In particular, $\gamma_j = 1$ implies $c_j = c_{0,j}$, whereas $\gamma_j = 0$ leaves c_j unconstrained by the expert partition. Since the last node is necessarily a changepoint, we set $c_p = c_{0,p} = \gamma_p = 1$.

Let

$$\gamma_{-j} = (\gamma_1, \dots, \gamma_{j-1}, \gamma_{j+1}, \dots, \gamma_p)$$

denote all components of γ except γ_j . We consider the two possible configurations

$$\gamma_j^{(1)} = (\gamma_1, \dots, \gamma_{j-1}, 1, \gamma_{j+1}, \dots, \gamma_p) \quad \text{and} \quad \gamma_j^{(0)} = (\gamma_1, \dots, \gamma_{j-1}, 0, \gamma_{j+1}, \dots, \gamma_p).$$

Conditional on the current ordered partition ρ'_p , the expert partition ρ_0 , and the remaining components of γ , Bayes' theorem gives

$$(21) \quad \Pr(\gamma_j = 1 \mid \rho'_p, \rho_0, \gamma_{-j}) = \frac{\Pr(\rho'_p \mid \rho_0, \gamma_j^{(1)}) \Pr(\gamma_j = 1)}{\Pr(\rho'_p \mid \rho_0, \gamma_j^{(1)}) \Pr(\gamma_j = 1) + \Pr(\rho'_p \mid \rho_0, \gamma_j^{(0)}) \Pr(\gamma_j = 0)} \\ \times C(\rho'_p \mid \rho_0, \gamma) A(\rho'_p),$$

where $C(\rho'_p \mid \rho_0, \gamma)$ and $A(\rho'_p)$ denote the compatibility and admissibility operators, respectively. Since $\gamma_j \sim \text{Bernoulli}(\eta)$, we have $\Pr(\gamma_j = 1) = \eta$ and $\Pr(\gamma_j = 0) = 1 - \eta$.

We enforce compatibility with the expert partition through two constraints. First, we set $\gamma_j = 0$ whenever $c_{0,j} = 0$, so that we update only the components $j \in \{i : c_{0,i} = 1\}$. Second, if $c_j = 0$ and $c_{0,j} = 1$, then $\Pr(\gamma_j = 1 \mid \rho'_p, \rho_0, \gamma_{-j}) = 0$, since setting $\gamma_j = 1$ would require $c_j = c_{0,j} = 1$ and would therefore be incompatible with the current partition. In this case, γ_j is deterministically set to zero.

Let

$$H = \sum_{j=1}^p \gamma_j$$

denote the current number of subgroups. Since $\gamma_p = 1$, the subgroups are obtained by considering the nodes lying between two consecutive entries of γ equal to one.

Suppose that $j \in \{i : c_{0,i} = 1\}$ is currently being updated. First consider $\gamma_j = 1$, that is, we force node j to be a changepoint node. Let h^- and h^+ denote the two subgroups immediately to the left and right of j , respectively, under this configuration. Equivalently, define

$$j^- = \max\{\ell < j : \gamma_\ell = 1\}, \quad j^+ = \min\{\ell > j : \gamma_\ell = 1\},$$

where $j^- = 0$ if no such endpoint exists. The two subgroups are then

$$\mathcal{I}_j^- = \{j^- + 1, \dots, j\}, \quad \mathcal{I}_j^+ = \{j + 1, \dots, j^+\}.$$

Let

$$\tilde{p}_{h^-} = |\mathcal{I}_j^-|, \quad \tilde{p}_{h^+} = |\mathcal{I}_j^+|$$

denote their respective numbers of nodes. Within each subgroup, the current changepoint vector $\mathbf{c}$ induces an ordered partition. Let K_{h^-} and K_{h^+} denote the corresponding numbers of clusters, and let

$$\mathbf{p}_{h^-} = (p_{h^-,1}, \dots, p_{h^-,K_{h^-}}), \quad \mathbf{p}_{h^+} = (p_{h^+,1}, \dots, p_{h^+,K_{h^+}})$$

denote the corresponding vectors of cluster cardinalities.

Conversely, under the configuration $\gamma_j = 0$, node j is not constrained by the expert partition and the two neighbouring subgroups are merged into a single subgroup, denoted by h^u . This subgroup contains

$$\mathcal{I}_j^u = \mathcal{I}_j^- \cup \mathcal{I}_j^+$$

and therefore has size

$$\tilde{p}_{h^u} = \tilde{p}_{h^-} + \tilde{p}_{h^+}.$$

Let K_{h^u} denote the number of clusters induced by $\mathbf{c}$ within this merged subgroup, and let

$$\mathbf{p}_{h^u} = (p_{h^u,1}, \dots, p_{h^u,K_{h^u}})$$

be the corresponding vector of cluster cardinalities.

Thus, changing γ_j affects only the local configuration (h^-, h^+, h^u) , while all other subgroups remain unchanged. This allows the full conditional of γ_j to be evaluated using only the corresponding local partition contributions.

Under the proposed informed ordered partition prior, we can write

$$\Pr(\rho'_p \mid \rho_0, \gamma) \propto \prod_{h=1}^H \mathcal{P}_h,$$

where

$$\mathcal{P}_h = \frac{1}{K_h!} \binom{\tilde{p}_h}{p_{h,1}, \dots, p_{h,K_h}} \Pi(p_{h,1}, \dots, p_{h,K_h})$$

denotes the contribution of subgroup h , including the EPPF term $\Pi(\cdot)$, the multinomial coefficient and the factorial correction associated with the ordered partition prior (see Equation (9)).

When $\gamma_j = 1$, the local contribution is $\mathcal{P}_{h^-} \mathcal{P}_{h^+}$, whereas when $\gamma_j = 0$, it is $\mathcal{P}_{h^u}$. All contributions associated with subgroups different from h^- , h^+ , and h^u are identical under the two configurations and therefore cancel in the ratio defining the full conditional. Consequently, Equation (21) simplifies to

$$(22) \quad \Pr(\gamma_j = 1 \mid \rho'_p, \rho_0, \gamma_{-j}) = \frac{\eta \mathcal{P}_{h^-} \mathcal{P}_{h^+}}{\eta \mathcal{P}_{h^-} \mathcal{P}_{h^+} + (1 - \eta) \mathcal{P}_{h^u}}.$$

Equivalently,

$$\Pr(\gamma_j = 1 \mid \rho'_p, \rho_0, \gamma_{-j}) = \left[1 + \frac{1 - \eta}{\eta} \frac{\mathcal{P}_{h^u}}{\mathcal{P}_{h^-} \mathcal{P}_{h^+}} \right]^{-1}.$$

Thus, the posterior odds in favour of enforcing the expert-defined changepoint at node j are obtained by multiplying the prior odds $\eta/(1 - \eta)$ by the ratio

$$\frac{\mathcal{P}_{h^-} \mathcal{P}_{h^+}}{\mathcal{P}_{h^u}},$$

which compares the prior contribution obtained by splitting the neighbourhood of j into two subgroups with that obtained by keeping the two sides within a single subgroup.

We use the ordered Poisson–Dirichlet EPPF within each subgroup. For a subgroup containing $\tilde{p}_h$ nodes and K_h clusters with cardinalities $(p_{h,1}, \dots, p_{h,K_h})$, satisfying $\sum_{k=1}^{K_h} p_{h,k} = \tilde{p}_h$, the corresponding ordered partition contribution is

$$(23) \quad \mathcal{P}_h = \frac{\tilde{p}_h!}{K_h!} \frac{\prod_{k=1}^{K_h-1} (\theta + k\sigma)}{(\theta + 1)^{\tilde{p}_h-1}} \prod_{k=1}^{K_h} \frac{(1 - \sigma)^{p_{h,k}-1}}{p_{h,k}!},$$

where $(a)_m = a(a+1) \cdots (a+m-1)$ denotes the rising factorial, with $(a)_0 = 1$, and $\sigma \in [0, 1)$ with $\theta > -\sigma$.

Plugging Equation (23) into Equation (22), we obtain

$$\begin{aligned} \Pr(\gamma_j = 1 \mid \rho'_p, \rho_0, \gamma_{-j}) &= \frac{\eta \mathcal{P}_{h^-} \mathcal{P}_{h^+}}{\eta \mathcal{P}_{h^-} \mathcal{P}_{h^+} + (1 - \eta) \mathcal{P}_{h^u}}, \\ \mathcal{P}_{h^-} &= \frac{\tilde{p}_{h^-}!}{K_{h^-}!} \frac{\prod_{k=1}^{K_{h^-}-1} (\theta + k\sigma)}{(\theta + 1)^{\tilde{p}_{h^-}-1}} \prod_{k=1}^{K_{h^-}} \frac{(1 - \sigma)_{p_{h^-,k}-1}}{p_{h^-,k}!}, \\ \mathcal{P}_{h^+} &= \frac{\tilde{p}_{h^+}!}{K_{h^+}!} \frac{\prod_{k=1}^{K_{h^+}-1} (\theta + k\sigma)}{(\theta + 1)^{\tilde{p}_{h^+}-1}} \prod_{k=1}^{K_{h^+}} \frac{(1 - \sigma)_{p_{h^+,k}-1}}{p_{h^+,k}!}, \\ \mathcal{P}_{h^u} &= \frac{\tilde{p}_{h^u}!}{K_{h^u}!} \frac{\prod_{k=1}^{K_{h^u}-1} (\theta + k\sigma)}{(\theta + 1)^{\tilde{p}_{h^u}-1}} \prod_{k=1}^{K_{h^u}} \frac{(1 - \sigma)_{p_{h^u,k}-1}}{p_{h^u,k}!}. \end{aligned} \tag{24}$$

This is the full conditional probability used to update each γ_j for which $c_{0,j} = 1$. Specifically, the MCMC algorithm draws

$$\gamma_j \mid \rho'_p, \rho_0, \gamma_{-j} \sim \text{Bernoulli}(q_j),$$

where q_j is given by Equation (24). For the compatibility constraint, if $c_{0,j} = 0$, γ_j is fixed to zero and no update is performed. Likewise, if $c_j = 0$ under the current partition, the value $\gamma_j = 1$ is incompatible with the current $\mathbf{c}$, and the update is deterministically set to $\gamma_j = 0$.

[Il seguente lo teniamo? L'ho messo ma forse non necessario.] For numerical stability, the three local contributions are evaluated on the logarithmic scale. In particular, defining

$$\ell_N = \log(\eta) + \log \mathcal{P}_{h^-} + \log \mathcal{P}_{h^+}, \quad \ell_D = \log(1 - \eta) + \log \mathcal{P}_{h^u},$$

the Bernoulli probability can equivalently be computed as

$$q_j = \frac{\exp(\ell_N)}{\exp(\ell_N) + \exp(\ell_D)} = \text{logit}^{-1}(\ell_N - \ell_D).$$

This local computation avoids evaluating the full prior probability of the partition at every update and requires only the evaluation of the three subgroup contributions affected by the proposed value of γ_j .

B.6. Update of the Random Partition. Conditional on the current value of the auxiliary vector γ , the ordered partition ρ'_p is updated separately within each of the H subgroups induced by γ . Recall that ρ'_p is represented by a vector of positive integers,

$$\rho'_p = (p_1, \dots, p_K), \quad \sum_{k=1}^K p_k = p,$$

where p_k denotes the number of consecutive nodes belonging to the k th group. The update is performed through local modifications of the partition, using two types of Metropolis-Hastings moves: a split/merge move, which changes the number of groups, and a shuffle move, which preserves the number of groups while modifying the allocation of nodes between two adjacent groups.

B.6.0.1. Split/merge moves within each subgroup. Let $\mathcal{I}_h$ denote the h th subgroup induced by γ , and let

$$\rho_h = (p_{h,1}, \dots, p_{h,K_h})$$

denote the restriction of the current partition to this subgroup. Thus, $p_{h,k}$ is the number of nodes in the k th group within subgroup h , K_h is the current number of groups, and

$$\tilde{p}_h = \sum_{k=1}^{K_h} p_{h,k}$$

is the number of nodes in the subgroup. The update is performed independently for $h = 1, \dots, H$. Equivalently, we can use the changepoint node representation and write c_h , $h = 1, \dots, H$, with $c_h = (c_{h,1}, \dots, c_{h,\tilde{p}_h})$ where $c_{h,j} \in \{0, 1\}$, $h = 1, \dots, H$, $j = 1, \dots, \tilde{p}_h$.

With probability p_{split} we propose a split move, whereas with probability $1 - p_{\text{split}}$ we propose a merge move. The move is switched whenever the selected type is not feasible: a split move cannot be proposed when $K_h = \tilde{p}_h$, since every node already forms a singleton group, while a merge move cannot be proposed when $K_h = 1$, since there is no internal changepoint to remove.

For a split move, a candidate node is selected among the nodes that are not currently changepoints of c_h . Splitting at the selected node increases the number of groups by one. Conversely, for a merge move, a candidate is selected among the current changepoints that can be removed, and the two groups adjacent to the selected changepoint are merged.

The space of possible configurations of c_h , for each h , can be large, so we preferentially propose changepoints in locations with high posterior probability, and delete them in regions with low posterior probability. We achieve this following the idea of [Benson and Friel \(2018\)](#), by introducing two p -dimensional iteration-specific vectors of unnormalized selection weights, $\mathbf{a}^{(t)} = (a_1^{(t)}, \dots, a_p^{(t)})^\top$ and $\mathbf{d}^{(t)} = (d_1^{(t)}, \dots, d_p^{(t)})^\top$, which we split into H subgroups so that, for $h = 1, \dots, H$, we can write

$$\mathbf{a}_h^{(t)} = (a_{h,1}^{(t)}, \dots, a_{h,\tilde{p}_h}^{(t)}), \quad \mathbf{d}_h^{(t)} = (d_{h,1}^{(t)}, \dots, d_{h,\tilde{p}_h}^{(t)}),$$

where $a_{h,j}^{(t)}$ and $d_{h,j}^{(t)}$ represent the unnormalized probabilities of choosing node j in sub-group h as a candidate for a split or merge move at iteration t , respectively. We define the normalizing sums as

$$A_h^{(t)} = \sum_{i:c_{h,i}=0} a_i^{(t)}, \quad D_h^{(t)} = \sum_{i:c_{h,i}=1} d_i^{(t)}.$$

Thus, if node j is not a changepoint ($c_{h,j} = 0$), a split move proposes adding a changepoint at j with probability $a_{h,j}^{(t)} / A_h^{(t)}$; if it is a changepoint ($c_{h,j} = 1$), a merge move proposes removing it with probability $d_{h,j}^{(t)} / D_h^{(t)}$. We impose $a_{H,p}^{(t)} = d_{H,p}^{(t)} = 0$ for all iterations t .

The weights are adaptive in the sense that we update the elements of the two weights vectors, $\mathbf{a}^{(t)}$ in case of a split move, and $\mathbf{d}^{(t)}$ in case of a merge move, at each iteration t . If a split move at node j has been accepted, then

$$\log(a_j^{(t+1)}) = \log(a_j^{(t)}) + \frac{b}{t/p} (\alpha_{\text{split}} - \alpha_{\text{target}}),$$

whereas if a merge move at node j has been accepted, then

$$\log(d_j^{(t+1)}) = \log(d_j^{(t)}) + \frac{b}{t/p} (\alpha_{\text{split}} - \alpha_{\text{target}}),$$

where $b > 0$ is an initial adaptation parameter and α_{target} is the target Metropolis-Hastings acceptance rate. We refer to [Benson and Friel \(2018\)](#) for further details.

[qua da cambiare un attimo come sto scrivendo la frase, è provvisoria giusto per capire] We now describe the acceptance probability of a split move. Suppose node j in subgroup h is selected to split its current cluster into two, with $c_{h,j} = 0$ and $c_{h,j}^* = 1$. Then we have $\mathbf{c}_h$ and $\mathbf{c}_h^*$ denoting the current and proposed partitions for subgroup h , respectively, and the whole vectors $\mathbf{c}$ and $\mathbf{c}^*$ representing the current and proposed partitions, respectively. The number of groups within subgroup h increases by one. Under the admissibility assumption, there is only one valid way to split the affected cluster: nodes with index $i \leq j$, with i and j in subgroup h , must belong to the first cluster, and those with $l > j$, with l and j in subgroup h , must belong to the second cluster. The proposed split move is accepted with probability

$$\alpha_{\text{split}} = \min(1, R_+),$$

where

$$R_+ = \frac{P(G | \mathbf{c}^*)}{P(G | \mathbf{c})} \frac{P(\mathbf{c}_h^* | \gamma, \theta, \sigma)}{P(\mathbf{c}_h | \gamma, \theta, \sigma)} \frac{q(\mathbf{c}_h | \mathbf{c}_h^*)}{q(\mathbf{c}_h^* | \mathbf{c}_h)}.$$

We refer to these three factors as the target ratio, the prior ratio (where we condition on the latent indicator γ and the hyperparameters of the Poisson-Dirichlet EPPF we use, σ and θ) and the proposal ratio.

Note that since a split move only splits one group into two adjacent groups, all terms in the prior ratio that do not involve the modified groups cancel, [teniamo o banale la giustificazione seguente?] this is why in the prior ratio and proposal ratio we keep only $\mathbf{c}_h^*$ and $\mathbf{c}_h$. Symmetrically, a merge move at node j can be viewed as the reverse of a split move, transitioning from the proposed partition $\mathbf{c}'$ back to the current partition $\mathbf{c}$. We therefore define the acceptance probability of a merge move as $\alpha_{\text{merge}} = \min(1, R_-)$.

Suppose that the selected changepoint splits a group of size $a + b$ into two groups of sizes a and b . If the current partition has K_h groups, the resulting prior ratio for the split move is

$$\frac{\Pr(\mathbf{c}_h^* | \gamma, \theta, \sigma)}{\Pr(\mathbf{c}_h | \gamma, \theta, \sigma)} = \frac{(a+b)!}{a!b!} \frac{\theta + K_h \sigma}{K_h + 1} \frac{(1-\sigma)_{a-1}(1-\sigma)_{b-1}}{(1-\sigma)_{a+b-1}}.$$

For a merge move, the inverse ratio is used. This expression follows directly from the Poisson-Dirichlet contribution to the informed ordered partition prior, since all unaffected groups cancel from the ratio.

For what concerns the proposal ratio, define the sets of available nodes for split or merge as

$$\mathcal{A}_h^{(a)} = \{j \mid c_{h,j} = 0\}, \quad \mathcal{A}_h^{(d)} = \{j \mid c_{h,j} = 1\},$$

with the convention that in the last sub-group $h = H$ the final node cannot be selected, so its weight is set to zero. We then have that

$$A_h = \sum_{j \in \mathcal{A}_h^{(a)}} a_{h,j}, \quad D_h = \sum_{j \in \mathcal{A}_h^{(d)}} d_{h,j}.$$

At each update, we first choose whether to attempt a split or merge with probabilities p_{split} and $p_{\text{merge}} = 1 - p_{\text{split}}$, respectively. Conditional on choosing a split move, the proposal ratio is

$$\frac{q(\mathbf{c}_h | \mathbf{c}_h^*)}{q(\mathbf{c}_h^* | \mathbf{c}_h)} = \frac{1 - p_{\text{split}}}{p_{\text{split}}} \frac{A_h}{a_{h,j}} \frac{d_{h,j}}{d_{h,j} + D_h},$$

where the first factor accounts for the relative probabilities of selecting a merge versus a split move, and the second and third factors compare the probability of choosing node j for the split (second) with the probability of selecting the same node for the reverse merge (third),

conditional on the split having been performed (that is why we must add $d_{h,j}$). Conditional on choosing a merge move we obtain

$$\frac{q(\mathbf{c}_h | \mathbf{c}_h^*)}{q(\mathbf{c}_h^* | \mathbf{c}_h)} = \frac{p_{\text{split}}}{1 - p_{\text{split}}} \frac{D_h}{d_{h,j}} \frac{a_{h,j}}{a_{h,j} + A_h}.$$

We must consider two special cases in which, within each sub-group h , either only one cluster is present (i.e., there are no changepoints and $K_h = 1$) or all nodes form separate clusters (i.e., every node is a changepoint and $K_h = \tilde{p}_h$). In these boundary cases, the proposal ratio reduces to

$$\frac{1 - p_{\text{add}}}{p_{\text{add}}} \frac{A_h}{a_{h,j}} \quad \text{or} \quad \frac{p_{\text{add}}}{1 - p_{\text{add}}} \frac{D_h}{d_{h,j}},$$

respectively.

We now turn to the target ratio. Although the split/merge proposal is performed within a single subgroup h , the graph likelihood $\Pr(G | \mathbf{c})$ is defined with respect to the complete partition $\mathbf{c}$. Consequently, the target ratio is computed using the full graph G and the full current and proposed partitions, while the modification itself affects only the two groups involved in the split or merge. For a given partition $\rho'_p = (p_1, \dots, p_K)$, let S_{uv} and S_{uv}^* denote, respectively, the number of edges and non-edges between groups u and v , where $1 \leq u \leq v \leq K$. Under the Beta-Bernoulli specification for the edges, the graph contribution to the target distribution can be written as

$$P(G | \rho'_p) = \prod_{u=1}^K \prod_{v=u}^K \frac{B(\alpha_1 + S_{uv}, \alpha_2 + S_{uv}^*)}{B(\alpha_1, \alpha_2)},$$

[Dobbiamo metterlo o basta definizione in main paper?] where $B(\cdot, \cdot)$ denotes the Beta function.

Consider first a split move within subgroup h . Let ρ'_p and ρ_p^{*} denote the current and proposed full partitions, respectively. If the split is performed on the s_h th group of subgroup h , the corresponding group in the full partition has index

$$s = \sum_{\ell=1}^{h-1} K_\ell + s_h.$$

Thus, the split changes group s in the current partition into two adjacent groups, s and $s + 1$, in the proposed partition. If the current partition contains K groups, the proposed partition therefore contains $K + 1$ groups.

Let

$$S_{uv}, \quad S_{uv}^*$$

denote the edge and non-edge counts computed from the current partition, and let

$$S'_{uv}, \quad S_{uv}^{*}$$

denote the corresponding quantities computed from the proposed partition. Because the split only modifies group s , all factors involving pairs of groups that are unaffected by the split

cancel between numerator and denominator. The resulting ratio can therefore be written as

$$\begin{aligned}
 (25) \quad \frac{P(G \mid \rho_p^*)}{P(G \mid \rho_p')} &= \left[\frac{1}{B(\alpha_1, \alpha_2)} \right]^{K+1} \\
 &\times \frac{B(\alpha_1 + S'_{ss}, \alpha_2 + S_{ss}^*) B(\alpha_1 + S'_{s,s+1}, \alpha_2 + S_{s,s+1}^*) B(\alpha_1 + S'_{s+1,s+1}, \alpha_2 + S_{s+1,s+1}^*)}{B(\alpha_1 + S_{ss}, \alpha_2 + S_{ss}^*)} \\
 &\times \prod_{\ell=1}^{s-1} \frac{B(\alpha_1 + S'_{\ell s}, \alpha_2 + S_{\ell s}^*) B(\alpha_1 + S'_{\ell,s+1}, \alpha_2 + S_{\ell,s+1}^*)}{B(\alpha_1 + S_{\ell s}, \alpha_2 + S_{\ell s}^*)} \\
 &\times \prod_{m=s+2}^{K+1} \frac{B(\alpha_1 + S'_{sm}, \alpha_2 + S_{sm}^*) B(\alpha_1 + S'_{s+1,m}, \alpha_2 + S_{s+1,m}^*)}{B(\alpha_1 + S_{s,m-1}, \alpha_2 + S_{s,m-1}^*)}.
 \end{aligned}$$

The first factor in (25) arises because the proposed partition contains one additional group and hence one additional Beta-normalizing factor. The second factor contains the three new pairwise contributions involving the two groups created by the split, divided by the contribution corresponding to the original group. The first product accounts for the interactions of the two new groups with all groups preceding group s . The second product accounts for their interactions with all groups following group s . Notice that the indices in the denominator of the latter product are shifted by one, since the split introduces a new group in the proposed partition.

The target ratio for a merge move is obtained by reversing the split move. In particular, if a merge combines groups s and $s+1$, the proposed partition contains one fewer group, and the corresponding ratio is simply the reciprocal of (25), evaluated with the current and proposed partitions interchanged.

B.6.0.2. Shuffle moves. The split/merge moves modify the number of groups within a subgroup, but they do not directly allow the sampler to redistribute the nodes between two existing adjacent groups while keeping the number of groups fixed. To complement these moves and improve mixing, we introduce a shuffle move (Corradin et al., 2022). This move is performed on the current full partition and preserves its number of groups. Let $\rho_p' = (p_1, \dots, p_K)$ denote the current partition. If $K \geq 2$, we first select uniformly an index $k \in \{1, \dots, K-1\}$ corresponding to two adjacent groups with sizes p_k and p_{k+1} . If both groups are singletons, i.e., $p_k = p_{k+1} = 1$, the shuffle move is not performed, since there is no different allocation of the nodes between the two groups that can be obtained without changing the partition. Otherwise, we propose a new size ℓ for the k th group by sampling from

$$\{1, \dots, p_k + p_{k+1} - 1\} \setminus \{p_k\},$$

where the current value p_k is excluded to ensure that the proposal corresponds to a different partition. The proposed partition ρ_p^* is then obtained by replacing the two selected group sizes according to

$$p_k^* = \ell, \quad p_{k+1}^* = p_k + p_{k+1} - \ell,$$

while leaving all other group sizes unchanged. Thus,

$$\rho_p^* = (p_1, \dots, p_{k-1}, \ell, p_k + p_{k+1} - \ell, p_{k+2}, \dots, p_K).$$

Note that the total number of nodes and the number of groups remain unchanged. The shuffle move is accepted with probability $\alpha_{\text{shuffle}} = \min(1, R_{\text{shuffle}})$, where

$$R_{\text{shuffle}} = \frac{\Pr(G \mid \rho_p^*) \Pr(\rho_p^* \mid \gamma, \theta, \sigma)}{\Pr(G \mid \rho_p') \Pr(\rho_p' \mid \gamma, \theta, \sigma)}.$$

The prior ratio for the shuffle move can be simplified as, on one hand, all terms in the Poisson-Dirichlet prior that do not involve the two modified groups cancel, and on the other hand the factors involving θ and the total number of groups cancel too as the move leaves K unchanged:

$$\frac{\Pr(\rho_p^* | \gamma, \theta, \sigma)}{\Pr(\rho_p' | \gamma, \theta, \sigma)} = \frac{p_k! p_{k+1}!}{\ell! (p_k + p_{k+1} - \ell)!} \frac{(1 - \sigma)_{\ell-1} (1 - \sigma)_{(p_k + p_{k+1} - \ell) - 1}}{(1 - \sigma)_{p_k-1} (1 - \sigma)_{p_{k+1}-1}}.$$

Consider now the graph likelihood contribution. Since the shuffle move does not change the number of groups, all normalizing factors $B(\alpha_1, \alpha_2)^{-1}$ cancel. Moreover, only terms involving groups k and $k + 1$ can differ between the current and proposed partitions. Consequently, the graph likelihood ratio can be written as

$$\begin{aligned} & \frac{P(G | \rho_p^*)}{P(G | \rho_p')} \\ &= \frac{B(\alpha_1 + S'_{kk}, \alpha_2 + S_{kk}^*) B(\alpha_1 + S'_{k,k+1}, \alpha_2 + S_{k,k+1}^*) B(\alpha_1 + S'_{k+1,k+1}, \alpha_2 + S_{k+1,k+1}^*)}{B(\alpha_1 + S_{kk}, \alpha_2 + S_{kk}^*) B(\alpha_1 + S_{k,k+1}, \alpha_2 + S_{k,k+1}^*) B(\alpha_1 + S_{k+1,k+1}, \alpha_2 + S_{k+1,k+1}^*)} \\ & \times \prod_{\ell=1}^{k-1} \frac{B(\alpha_1 + S'_{\ell k}, \alpha_2 + S_{\ell k}^*) B(\alpha_1 + S'_{\ell,k+1}, \alpha_2 + S_{\ell,k+1}^*)}{B(\alpha_1 + S_{\ell k}, \alpha_2 + S_{\ell k}^*) B(\alpha_1 + S_{\ell,k+1}, \alpha_2 + S_{\ell,k+1}^*)} \\ & \times \prod_{\ell=k+2}^K \frac{B(\alpha_1 + S'_{\ell k}, \alpha_2 + S_{\ell k}^*) B(\alpha_1 + S'_{\ell,k+1}, \alpha_2 + S_{\ell,k+1}^*)}{B(\alpha_1 + S_{\ell k}, \alpha_2 + S_{\ell k}^*) B(\alpha_1 + S_{\ell,k+1}, \alpha_2 + S_{\ell,k+1}^*)}. \end{aligned}$$

As for the split/merge moves, these quantities are computed using the whole graph G and the complete current and proposed partitions.

There is no proposal ratio in α_{shuffle} because the proposal is symmetric: the same mechanism used to redistribute the $p_k + p_{k+1}$ nodes between the two adjacent groups can be applied in reverse to recover the current partition from the proposed one.

B.7. Update of the EPPF Hyperparameters. Here we turn attention to the hyperparameters σ and θ proposed in [Martínez and Mena \(2014\)](#). We defined the corresponding prior distributions in Section 4.

The full conditional distribution of σ is given by

$$p(\sigma | \theta, p_1, \dots, p_K) \propto \sigma^{a-1} (1 - \sigma)^{b-1} (\theta + \sigma)^{c-1} e^{-d\sigma} \prod_{m=1}^{K-1} (\theta + m\sigma) \prod_{m=1}^K (1 - \sigma)_{(p_m-1)},$$

with $\sigma \in (\max\{-\theta, 0\}, 1)$. We employ an adaptive Metropolis-Hastings step to sample from this distribution.

For the posterior distribution of θ , we rely on the augmentation technique defined in Proposition 1 of [Martínez and Mena \(2014\)](#). This technique consists of introducing two latent variables, $\xi \sim \text{Gamma}(1, \theta + 1)$ and $\zeta \sim \text{Beta}(\theta + 2, p)$ so that the full conditional distribution of θ can be expressed as a mixture of $(-\sigma)$ -shifted gamma distributions. Namely,

$$p(\theta | \sigma, \xi, \zeta, K) = \sum_{j=0}^{K+1} w_j \text{Gamma}_{-\sigma}(c + j, d + \xi - \log(\zeta)),$$

where the mixing weights w_j , for $j = 0, \dots, K + 1$, are proportional to

$$w_j \propto \frac{\Gamma(c + j)}{\{\sigma(d + \xi - \log(\zeta))\}^j} \times \left[(p - \sigma)(p - \sigma + 1) \left| s_{K-1}^j \right| + (2p - 2\sigma + 1) \left| s_{K-1}^{j-1} \right| + \sigma^2 \left| s_{K-1}^{j-2} \right| \right],$$

where $|s_n^j|$ denotes the signless Stirling number of the first kind; see [Charalambides \(2002\)](#).

In particular, we recall that $|s_0^0| = 1$ and $|s_n^j| = 0$ for any $j \leq 0$ or $j > n$.

APPENDIX C: SIMULATION STUDY: ADDITIONAL MATERIAL

In this Section we provide additional visualizations for the simulation study presented in the main manuscript (Section 6). For each of the three simulated scenarios, we report the empirical distributions of the F1 score of the estimated graphs and of the Wasserstein distance between the true and estimated precision matrices, for both $n = 200$ and $n = 500$. We also report the empirical distributions of the ARI for the two scenarios with misspecified expert partitions, computed between the expert-defined partition and the estimated partition. These results further illustrate the robustness of the proposed approach to misspecified expert information: even for larger values of η , the ARI does not converge to one, indicating that the model can depart from an incorrectly specified expert partition despite assigning substantial prior weight to the expert information.

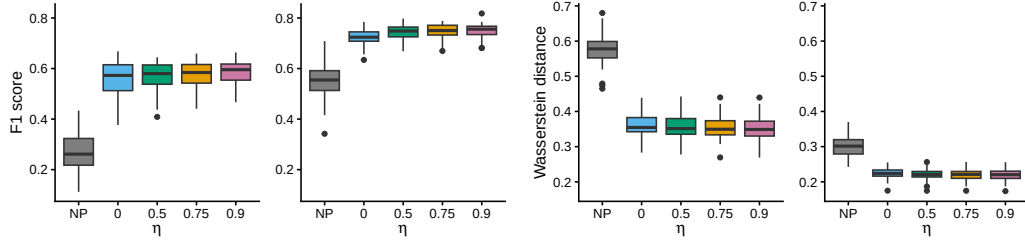


Fig 9: $\rho_0 = \rho_{\text{true}}$ setting. On the left panel, the empirical distribution of the F1 score of the estimated graphs. On the right panels, the empirical distribution of the Wasserstein distance between the true and estimated precision matrices. For each quantity, the left and right sub-panels correspond to $n = 200$ and $n = 500$, respectively.

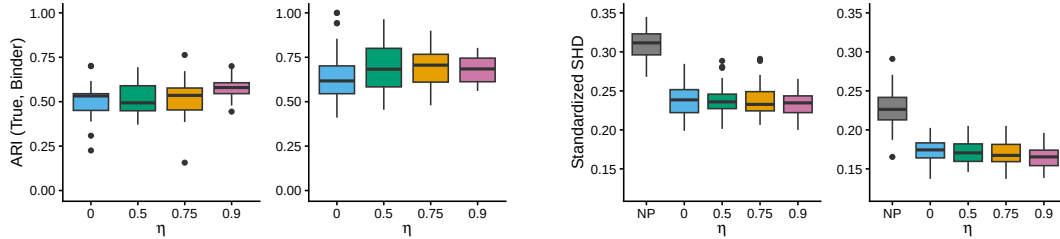


Fig 10: $\rho_0 = \rho_{\text{shift}}$ setting. On the left panel, the empirical distribution of the F1 score of the estimated graphs. On the right panels, the empirical distribution of the Wasserstein distance between the true and estimated precision matrices. For each quantity, the left and right sub-panels correspond to $n = 200$ and $n = 500$, respectively.

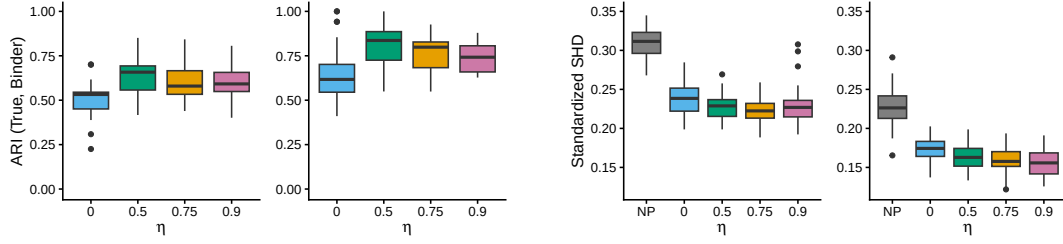


Fig 11: $\rho_0 = \rho_{\text{sm}}$ setting. On the left panel, the empirical distribution of the F1 score of the estimated graphs. On the right panels, the empirical distribution of the Wasserstein distance between the true and estimated precision matrices. For each quantity, the left and right sub-panels correspond to $n = 200$ and $n = 500$, respectively.

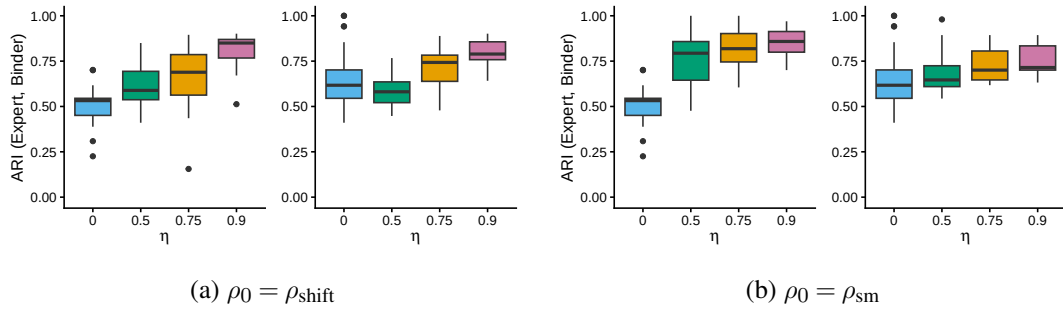


Fig 12: ARI between the posterior estimate of ρ and the corresponding informed partition. In each sub-panel, the left plot corresponds to $n = 200$ and the right plot to $n = 500$.

APPENDIX D: DISTANCE BETWEEN NON-STRAWBERRY AND STRAWBERRY COEFFICIENTS

We recall that the squared 2-Wasserstein distance between two laws $\mathcal{N}_p(\boldsymbol{\mu}_s, \boldsymbol{\Sigma}_s)$ and $\mathcal{N}_p(\boldsymbol{\mu}_{ns}, \boldsymbol{\Sigma}_{ns})$ ($\boldsymbol{\Sigma}_d = \boldsymbol{\Omega}_d^{-1}$, $d \in \{ns, s\}$) admits a closed-form expression combining a location component and a covariance component as expressed in Equation (14). It can be separated into a mean component $\|\boldsymbol{\mu}_s - \boldsymbol{\mu}_{ns}\|^2$ and a structure component $\text{tr}\left(\boldsymbol{\Sigma}_s + \boldsymbol{\Sigma}_{ns} - 2(\boldsymbol{\Sigma}_s^{1/2} \boldsymbol{\Sigma}_{ns} \boldsymbol{\Sigma}_s^{1/2})^{1/2}\right)$. The first term captures differences in location, while the second term quantifies discrepancies in the covariance structure. We estimate the distance via Monte Carlo using posterior draws for each $\eta \in \{0, 0.5, 0.75, 0.9\}$. Let $\{(\boldsymbol{\mu}_s^{(t)}, \boldsymbol{\Omega}_s^{(t)})\}_{t=1}^T$ and $\{(\boldsymbol{\mu}_{ns}^{(t')}, \boldsymbol{\Omega}_a^{(t')})\}_{t'=1}^T$ denote samples from the two posterior chains after burn-in, where the indices t and t' are sampled independently with replacement from $\{1, \dots, T\}$. We independently sample pairs (t, t') and compute

$$W_2^2((\boldsymbol{\mu}_s^{(s)}, \boldsymbol{\Omega}_s^{(t)}), (\boldsymbol{\mu}_{ns}^{(t')}, \boldsymbol{\Omega}_a^{(t')})), \quad t, t' = 1, \dots, T.$$

Table 1 reports summaries of the empirical distributions of the Wasserstein distances between the posterior distributions. The entries are defined as follows:

- **W_mean, W_q05, W_median, W_q95**: empirical mean, 5% quantile, median, and 95% quantile of the Monte Carlo samples of the Wasserstein distance.
- **W_prop_mean, W_prop_structure**: proportion of the total Wasserstein distance explained by the location and covariance components.

TABLE 1
Summary of Wasserstein distance between strawberry and non-strawberry posteriors.

	$\eta = 0$	$\eta = 0.5$	$\eta = 0.75$	$\eta = 0.9$
W_mean	0.519	0.513	0.518	0.507
W_q05	0.491	0.484	0.491	0.478
W_q50	0.519	0.513	0.517	0.507
W_q95	0.548	0.543	0.547	0.537
W_prop_mean	0.091	0.094	0.092	0.096
W_prop_structure	0.909	0.906	0.908	0.904

The results are remarkably stable across all values considered η . The average Wasserstein distance between the strawberry and non-strawberry posterior distributions remains close to 0.51, and the corresponding empirical quantiles exhibit negligible variation. This indicates that the inferred discrepancy between the two groups is largely insensitive to the strength of the prior information in the experts’ defined partition encoded through η . The Wasserstein decomposition reveals that the difference between the two distributions is driven primarily by changes in covariance structure. In proportional terms, only about 9%–10% of the total Wasserstein distance is explained by differences in the posterior means of β , while more than 90% is attributable to differences in the covariance structure. These findings are consistent with the graphical analyses presented earlier. While the posterior means of the latent coefficients are similar for strawberry and non-strawberry purees, the dependence structure encoded by the corresponding precision matrices differs substantially. Hence, the main distinction between the two groups lies in conditional dependence among the latent coefficients. Furthermore, the stability of the Wasserstein summaries across all values of η indicates that this conclusion is robust to the amount of prior information incorporated through the informed ordered partition model.

APPENDIX E: MORE ON STRAWBERRY AND NON-STRAWBERRY RESULTS

[Ho scritto qualcosa sui number of groups, che però forse non mettiamo nemmeno, decidere questa cosa]

This Section provides additional posterior results, visualizations, and comparisons across different values of the prior parameter η .

Figures 13 and 17 report the posterior probabilities of a changepoint at each of the $p = 40$ nodes for $\eta \in 0, 0.5, 0.75, 0.9$, for the non-strawberry and strawberry curves, respectively. Across specifications, the estimated partitions are broadly consistent with the structure elicited by the experts, although comparing across values of η illustrates the effect of incorporating expert information. In the uninformed case ($\eta = 0$), the posterior probability of a changepoint is often negligible over the central-to-right portion of the spectrum. As η increases, these probabilities tend to increase around changepoints identified by the experts, indicating that the informed prior encourages the model to recover structures supported by expert knowledge in regions where the data alone provide weaker evidence. Importantly, however, the model does not simply reproduce the expert-elicited partition. Some expert-identified changepoints receive limited posterior support, particularly for the non-strawberry curves toward the end of the spectral range. In these regions, the model instead identifies alternative changepoints that were not anticipated by the experts and assigns them relatively high posterior probability. For the strawberry curves, the agreement is generally stronger, with fewer deviations from the expert partition. The most notable difference occurs for $\eta = 0$, as expected in the absence of expert information. A peculiar behavior, also observed in the main manuscript, is that two adjacent

nodes are identified as changepoints for the non-strawberry curves when $\eta = 0.5$ and for the strawberry curves when $\eta = 0.9$. [teniamo o evitiamo? Io forse toglierei]Figures 14 and 18 report the posterior distributions of the number of groups. For all values of η , essentially no posterior mass is assigned to a single group, which corresponds to the special case in which the proposed model reduces to the no-partition model of Codazzi et al. (2022). This provides evidence in favor of a non-trivial latent block structure in the underlying dependence network. For the non-strawberry curves, the posterior distribution of the number of groups is asymmetric, with a mode at 6 for $\eta = 0$, at 9 for $\eta \in 0.5, 0.75$, and at 12 for $\eta = 0.9$, while substantial posterior mass is also assigned to nearby values. For the strawberry curves, the posterior distribution is generally shifted toward smaller numbers of groups. To close the comment about the partition analysis, Table 2 reports the ARI for increasing values of η . The ARI remains relatively stable across specifications, ranging from 0.54 to 0.65, indicating a moderate degree of agreement between the estimated and expert-elicited partitions, while also highlighting differences between the two.

Figures 15 and 19 report the posterior probabilities of edge inclusion. These probabilities are generally higher for the non-strawberry curves than for the strawberry curves, indicating stronger conditional dependence among the former, consistently with the results reported in the main manuscript. Figures 16 and 20 show the estimated graphs together with the corresponding estimated partitions. The difference in conditional dependence structure between the two classes is evident, with the non-strawberry curves yielding substantially denser graphs. The estimated graphs also exhibit some changes as η varies. For the non-strawberry curves, when $\eta = 0$, the sixth partition (i.e., the penultimate block) forms a clearly defined cluster, characterized by relatively strong within-block connectivity and connections to the second block. The connection between these two blocks persists as η increases, whereas the within-block connections of the sixth partition become progressively weaker. This indicates that increasing the strength assigned to expert information modifies the posterior dependence structure within this block, reducing the evidence for the connectivity identified under the uninformed specification. For the strawberry curves, the effect of η is less pronounced, although some differences are still apparent. In particular, increasing η results in somewhat stronger connectivity between the second and third groups and the subsequent groups, suggesting that incorporating expert information can reveal dependence structures that are less evident under the uninformed model. Finally, Table 3 reports the posterior graph densities for the different values of η . The non-strawberry graphs are consistently denser than the strawberry graphs, further confirming the stronger conditional dependence structure observed for the non-strawberry curves.

TABLE 2
ARI between the partitions estimated for strawberry and non-strawberry samples for $\eta \in \{0, 0.5, 0.75, 0.9\}$.

η	0	0.5	0.75	0.9
ARI	0.633	0.536	0.648	0.648

TABLE 3

Posterior mean graph density under different values of the hyperparameter η for the non-strawberry and the strawberry purees.

η	non-strawberry	Strawberry
0.00	0.58	0.18
0.50	0.59	0.12
0.75	0.58	0.12
0.90	0.62	0.22

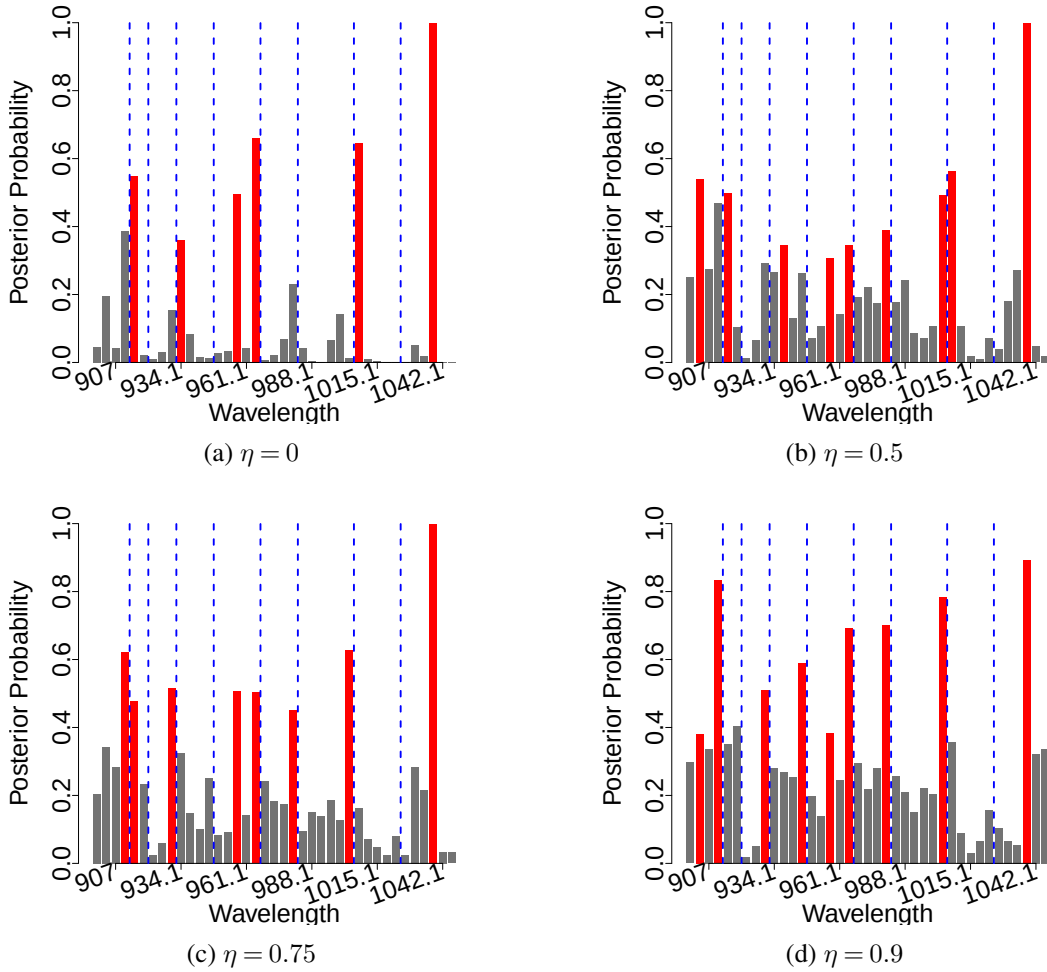


Fig 13: Posterior probabilities that each node corresponds to a changepoint for the non-strawberry purees, under different values of η . The estimated partition is indicated by red bars, while the expert-informed partition is indicated by a blue dotted line.

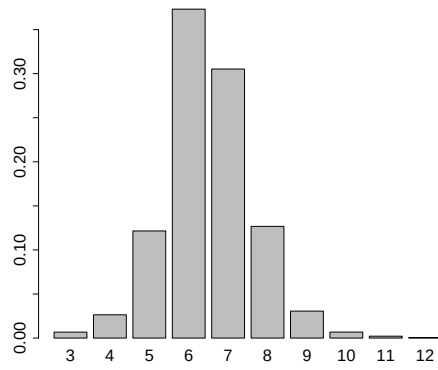
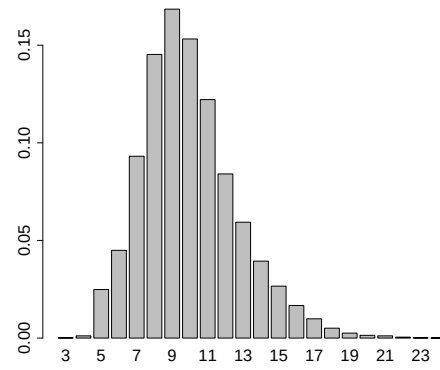
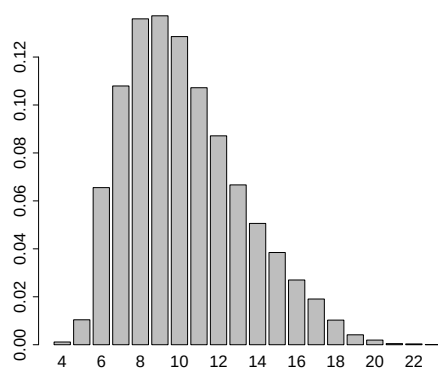
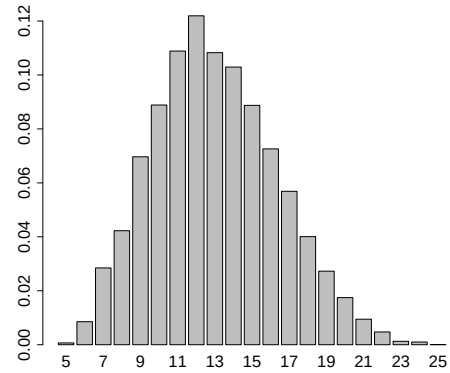
(a) $\eta = 0$ (b) $\eta = 0.5$ (c) $\eta = 0.75$ (d) $\eta = 0.9$

Fig 14: Posterior distribution of the number of groups for the non-strawberry purees under different values of η .

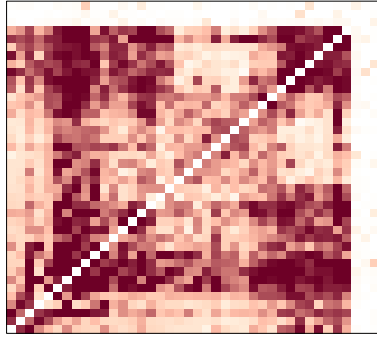
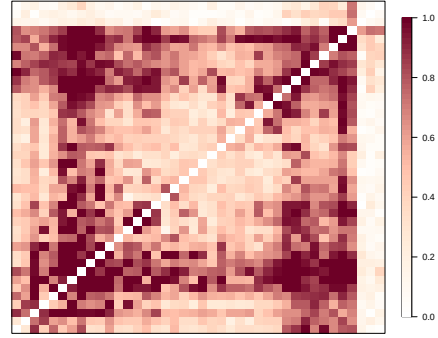
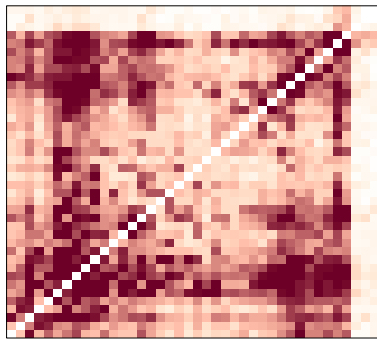
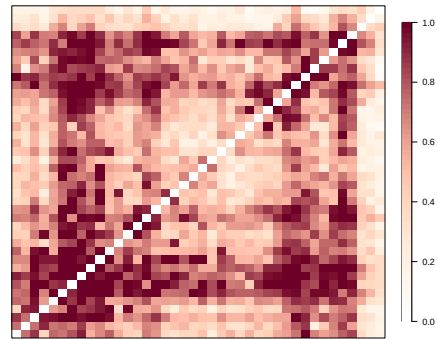
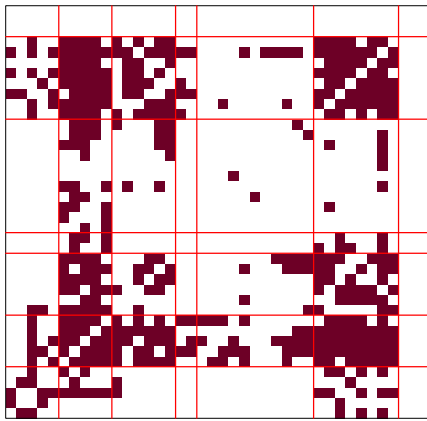
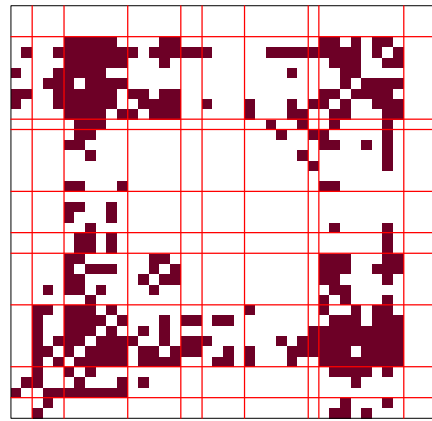
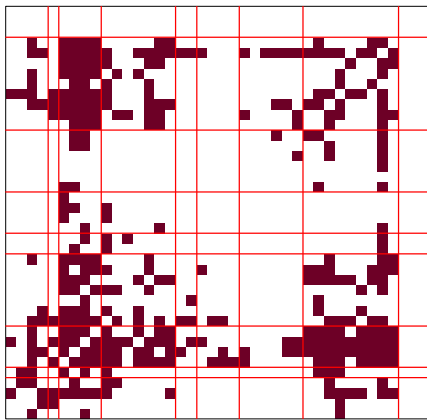
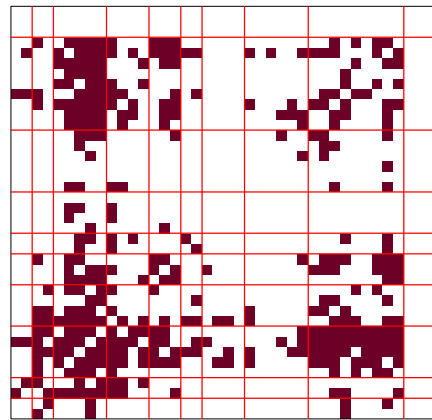
(a) $\eta = 0$ (b) $\eta = 0.5$ (c) $\eta = 0.75$ (d) $\eta = 0.9$

Fig 15: Posterior probabilities of edge inclusion for the non-strawberry purees under different values of η .

(a) $\eta = 0$ (b) $\eta = 0.5$ (c) $\eta = 0.75$ (d) $\eta = 0.9$ Fig 16: Estimated graphs for the non-strawberry purees under different values of η .

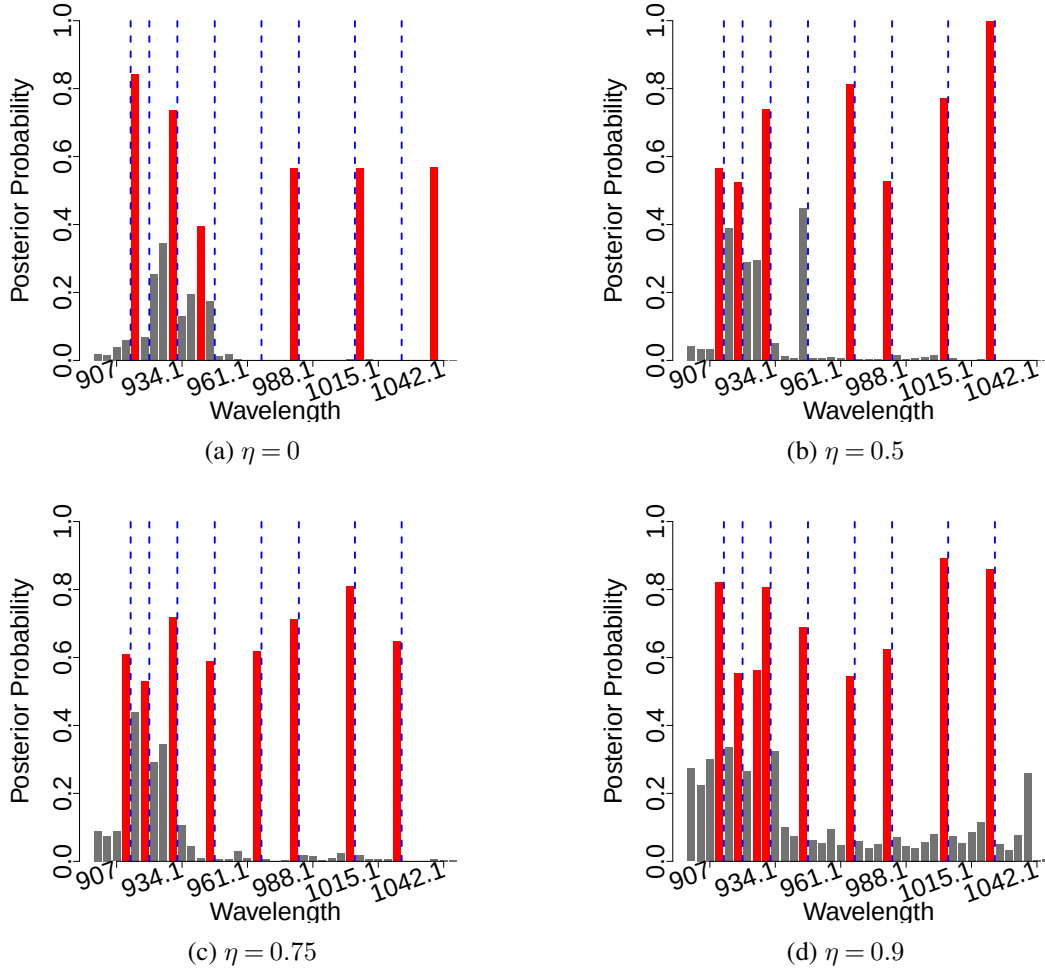


Fig 17: Posterior probabilities that each node corresponds to a changepoint for the strawberry purees, under different values of η . The estimated partition is indicated by red bars, while the expert-informed partition is indicated by a blue dotted line.

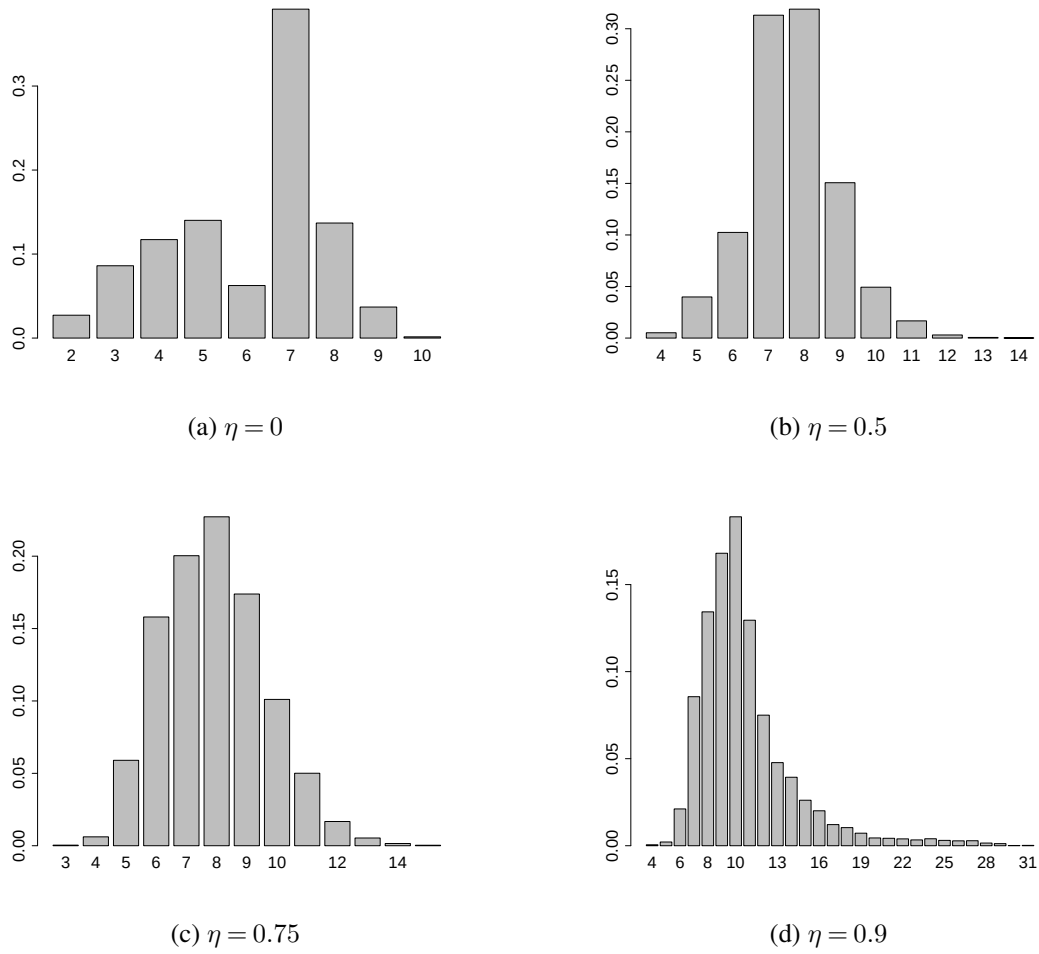


Fig 18: Posterior distribution of the number of groups for the strawberry purees under different values of η .

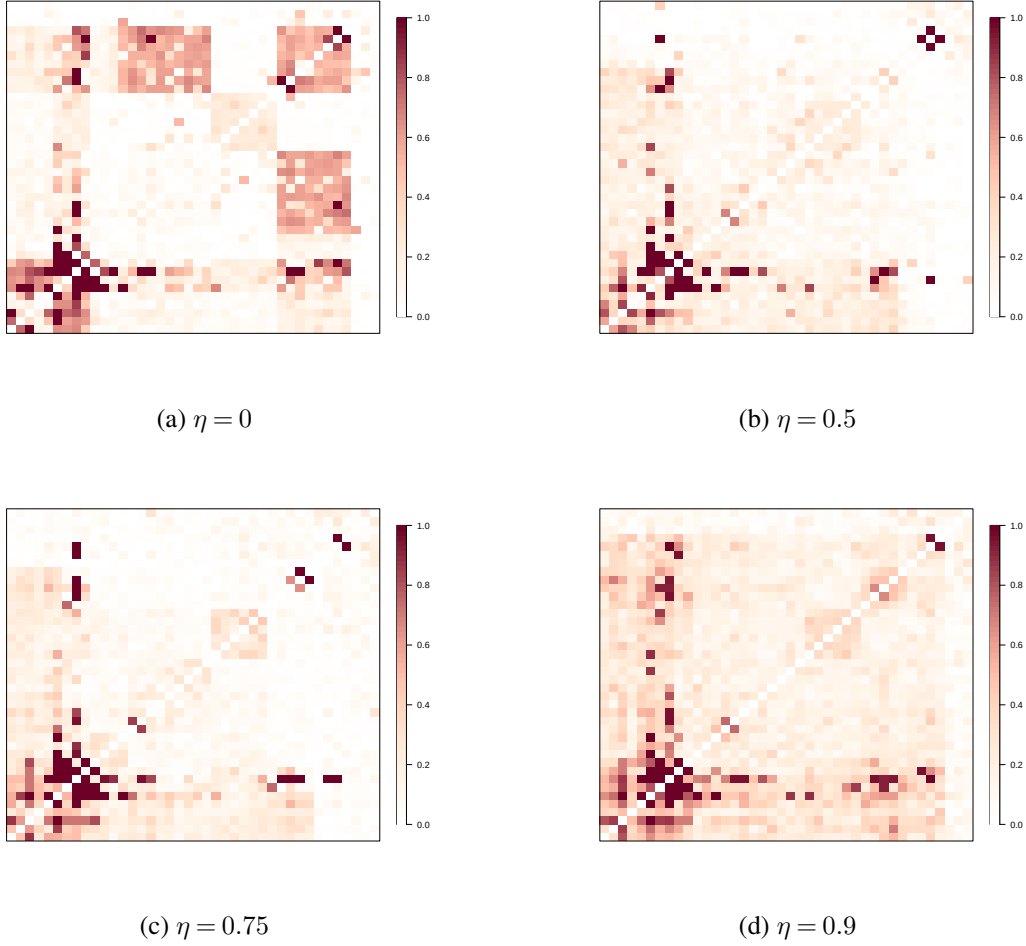
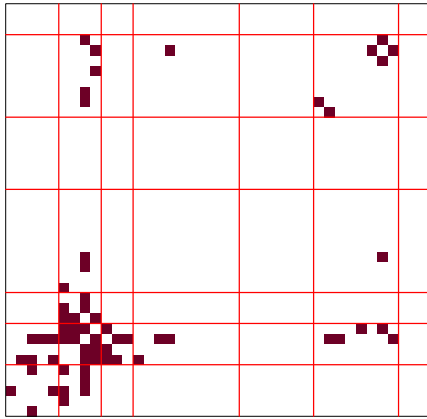
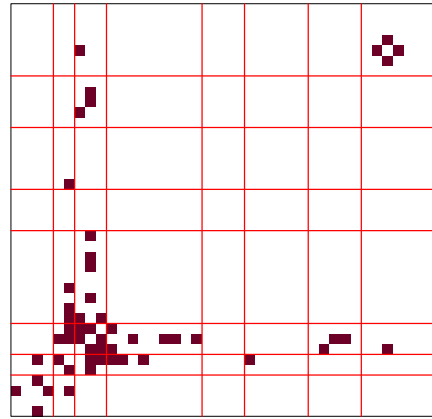
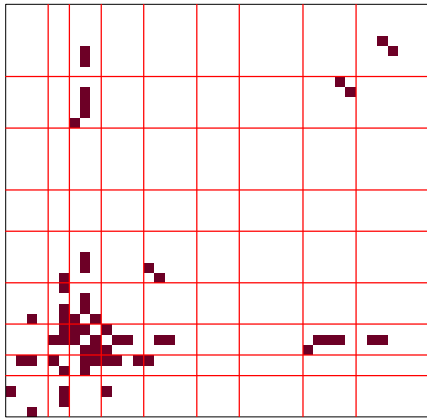
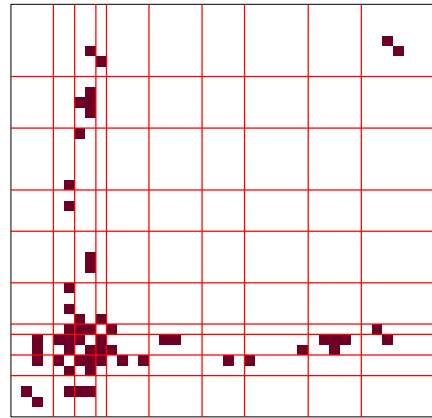


Fig 19: Posterior probabilities of edge inclusion for the strawberry purees under different values of η .

(a) $\eta = 0$ (b) $\eta = 0.5$ (c) $\eta = 0.75$ (d) $\eta = 0.9$ Fig 20: Estimated graphs for the strawberry purees under different values of η .

APPENDIX F: CLASSIFICATION: ADDITIONAL MATERIAL

Here we report confusion matrices and additional visualization for the classification task explained in Section 7.3.

Tables from 4 to 7 report the confusion matrices for all considered values of η , as well as under the baseline model. We achieved very high predictive accuracy, ranging between 96.3% and 96.7%. In all cases, only a very small number of curves were misclassified. In particular, the proposed methodology exhibited extremely high sensitivity toward the strawberry class, with only one misclassified strawberry puree under every considered specification.

Figures from 21a to 21d display the relationship between the Monte Carlo estimate $\hat{p}_s$ and the corresponding difference between the two class-specific utilities, for each η and the baseline model. The plots reveal a clear separation between strawberry and non-strawberry purees.

TABLE 4

Confusion matrix for $\eta = 0.5$. The selected threshold is $\zeta = 0.89$, yielding a classification accuracy of 0.967.

True/Predicted	Non-Strawberry	Strawberry
Non-Strawberry	141	9
Strawberry	1	149

TABLE 5

Confusion matrix for $\eta = 0.75$. The selected threshold is $\zeta = 0.86$, yielding a classification accuracy of 0.963.

True/Predicted	Non-Strawberry	Strawberry
Non-Strawberry	140	10
Strawberry	1	149

TABLE 6

Confusion matrix for $\eta = 0.9$. The selected threshold is $\zeta = 0.88$, yielding a classification accuracy of 0.967.

True/Predicted	Non-Strawberry	Strawberry
Non-Strawberry	141	9
Strawberry	1	149

TABLE 7

Confusion matrix for the baseline model. The selected threshold is $\zeta = 0.85$, yielding a classification accuracy of 0.963.

True/Predicted	Non-Strawberry	Strawberry
Non-Strawberry	140	10
Strawberry	1	149

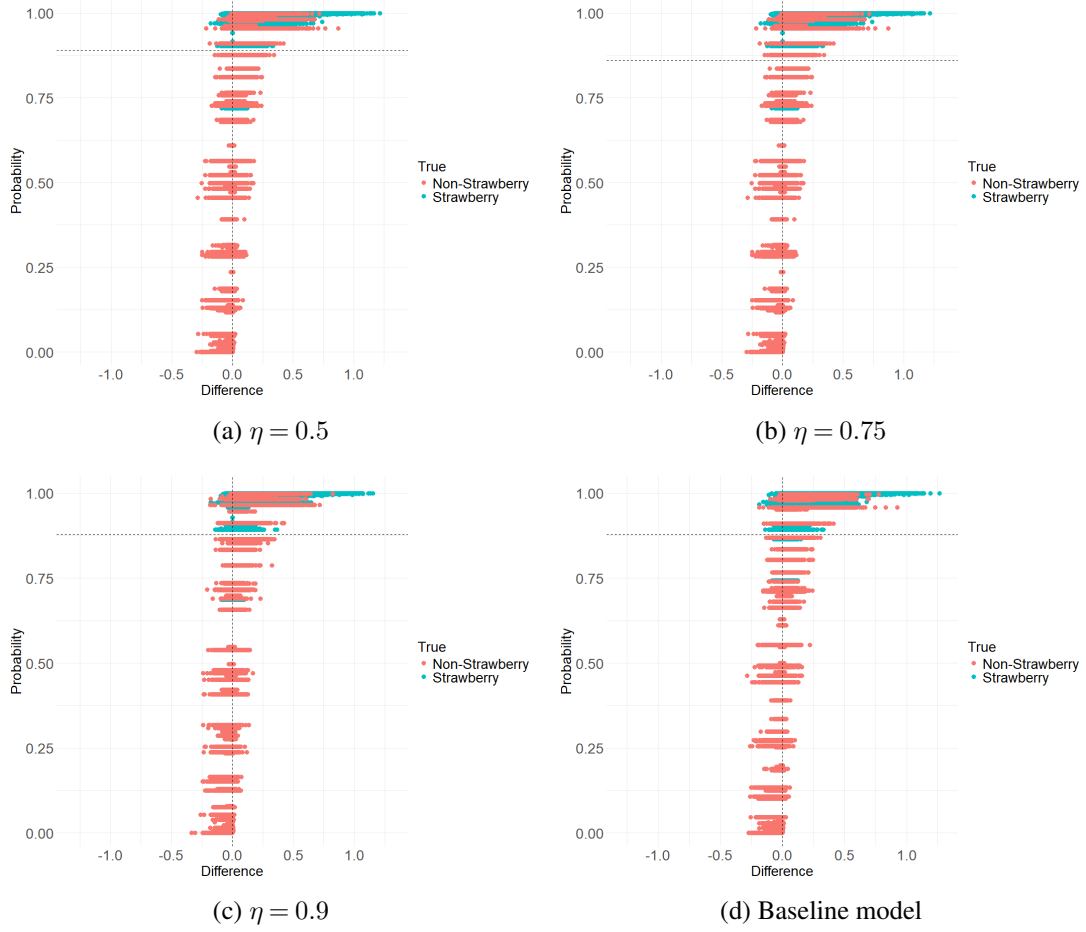


Fig 21: Utility difference versus estimated posterior probability $\hat{p}_s$ under different values of η and under the baseline model without partition information on the graph.

APPENDIX G: EXAMPLE OF INFORMED PARTITION

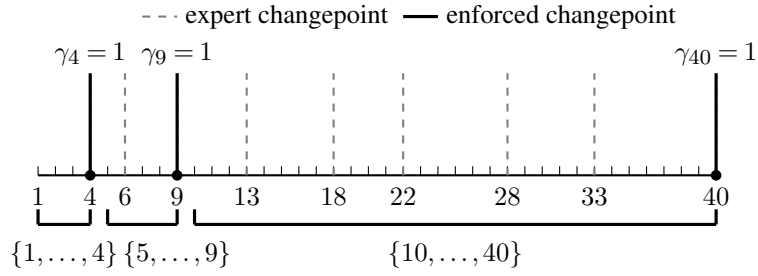


Fig 22: Expert-defined changepoints and the resulting admissible subgroups for $\gamma_4 = \gamma_9 = \gamma_{40} = 1$. Dashed lines indicate changepoints specified by the experts, whereas thick lines indicate those enforced by γ . Inside each subgroup, the model can find additional changepoints.